GIS

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Chapter 1

Introduction

SPATIAL THINKING

- **GIS = Geographic Information Systems**

✓ A system to organize, analyze, visualize, and share spatial data.

✓ Answers questions related to "where" things are or "where" things happen.

✓ (Where is the nearest hospital? Where does urbanization affect biodiversity?)

✓ Integrates spatial (location-based) data and attribute (descriptive) data.

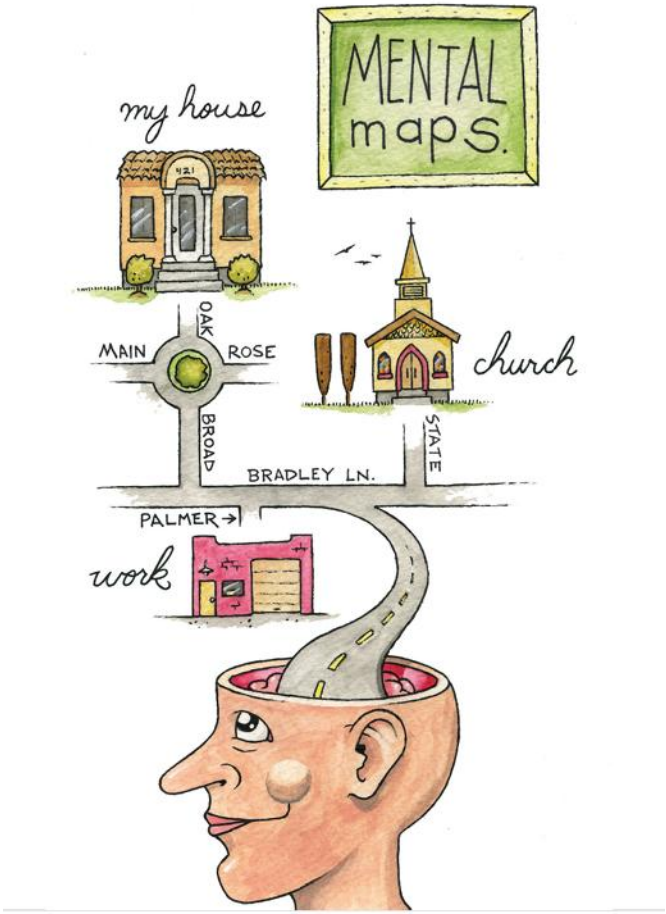
SPATIAL THINKING

- We use spatial thinking daily to understand and navigate our surroundings.
- Spatial thinking helps us answer **"Where?"** questions about objects and events.
- For instance: when you want to meet your friend at university you probably call him and ask where exactly you are and he answers I'm in front of central library. It's spatial thinking because it's related to locations.

✓ Geographic Information Systems (GIS) enhance spatial thinking by organizing and analyzing spatial data.

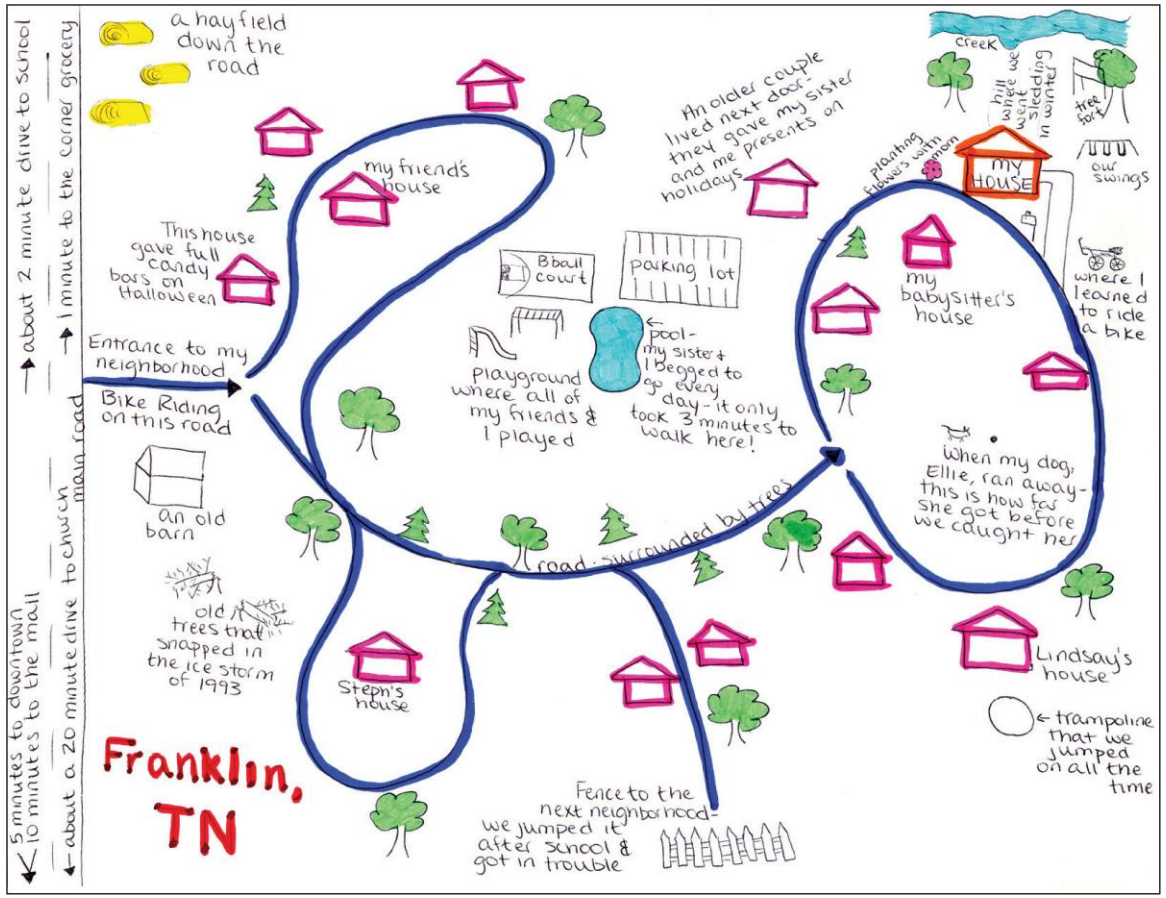
MENTAL MAPS

- A mental map is the representation of our environment stored in our brains.
- We use mental maps to:
  - ✓ Navigate cities and buildings.
  - ✓ Plan travel routes.
  - ✓ Understand geographic relationships.



MENTAL MAPS

- Example:
  - Assume your new neighbour asks you where the nearest grocery store is?
  - you answer go straight ahead for 200 meters, then turn left. Pass 3 blocks. It's at your left side next to the school.
  - In fact, you gave him this address using your mental map of your neighbourhood.



MENTAL MAPS

- **Mental Map Exercise**

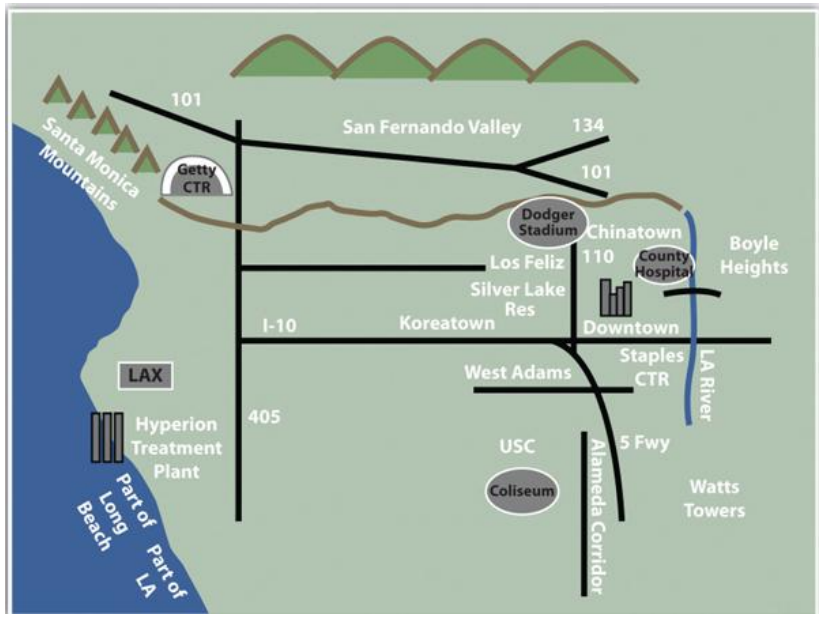
➢ Imagine a friend is visiting your city (your hometown) for the first time.

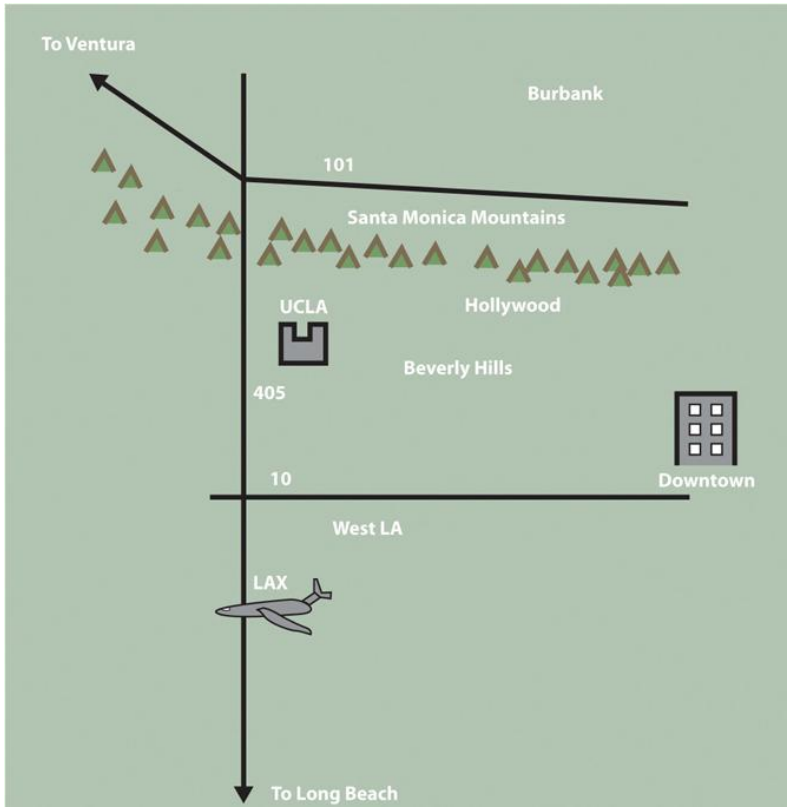
✓ Take a blank sheet of paper and draw a rough map of your city from memory.

✓ What landmarks do you include? What details do you omit?

MENTAL MAPS

- **Comparing Mental Maps**
- Different people's mental maps of the same place can vary significantly.
- Some factors that can influencing mental maps are:
  - ✓ Personal experiences.
  - ✓ Daily routines.
  - ✓ Cultural background.





GEOGRAPHIC QUESTIONS

- we use a lot of geographic knowledge in our daily life and we ask many questions that related to geography as well.
- For example: when you ask where is the nearest hospital it's a geographic question.
- In today's life answer to this question may not be so easy (at least in big cities) and also all geographic questions are not that simple.
- GIS helps us to have right understanding of our surrounding area and also can help us giving best answer to geographic questions.

GEOGRAPHIC QUESTIONS

- GIS helps answer **five key geographic questions:**

1. **Geographic Location** – Where is it?
2. **Geographic Distribution** – How is it spread?
3. **Geographic Association** – What is near it?
4. **Geographic Interaction** – How does it connect to other places?
5. **Geographic Change** – How has it changed over time?

GEOGRAPHIC QUESTIONS

Asking Good Geographic Questions

- Clear and well-structured questions lead to better GIS analysis. For example :
  - ✓ Poor question: *Why are there many hospitals?*
  - ✓ Better question: *Where are hospitals located in relation to population density?*

GEOGRAPHIC QUESTIONS

Practice: for below poor questions give some better questions instead.

1. **Poor:** Where are people living?
2. **Poor:** Why do floods happen?
3. **Poor:** Is there traffic in the city?
4. **Poor:** how is air quality in this city?
5. **Poor:** Are there many parks?

GEOGRAPHIC QUESTIONS

Answer - please note that your answers may be different

1. **Poor:** Where are people living?
   Better: Where are the most populated areas in the city?
2. **Poor:** Why do floods happen?
   Better: Where do floods happen most often, and why?
3. **Poor:** Is there traffic in the city?
   Better: Which roads have the most traffic during rush hours?
4. **Poor:** how is air quality in this city?
   Better: Which areas have the most numbers of days with polluted air during a month?
5. **Poor:** Are there many parks?
   Better: Where are parks located, and how close are they to neighborhoods?

REAL-WORLD APPLICATIONS OF SPATIAL THINKING

- **Urban Planning:** Where should new roads or buildings be constructed?
- **Disaster Management:** Where is the risk of flooding highest?
- **Marketing:** Where should a company open a new store?

✓ **We can answer all of these questions using GIS**

- Spatial thinking is a key skill for understanding and solving geographic problems.
- Mental maps help us visualize locations and relationships.
- Asking clear geographic questions improves our ability to use GIS effectively.

GEOGRAPHIC CONCEPTS

- Geographic concepts are important ideas that help us understand location, space, and movement on the Earth.
- These concepts are the foundation of Geographic Information Systems (GIS).

Here we have the most important GIS concepts:

1. Location
2. Direction
3. Distance
4. Space
5. Navigation

GEOGRAPHIC CONCEPTS

Location

- **Location** means *where something is on Earth*.
- It helps us find places and understand their position.

Types of Location

1. **Nominal Location:**
   - ✓ Uses names of places.
   - ✓ Example: Paris, Amazon Rainforest, Mount Everest.
2. **Relative Location:**
   - ✓ Describes a place compared to another place.
   - ✓ Example: *The school is near the bus station.*
   - ✓ This helps us understand where places are in relation to each other.
3. **Absolute Location :**
   - ✓ Gives an exact position using coordinates.
   - ✓ Example: *Eiffel Tower is at 48.8584° N, 2.2945° E.*
   - ✓ Nowdays, we use GPS (Global Positioning System) to determine absolute location. it uses satellites to find exact locations.

GEOGRAPHIC CONCEPTS

Direction

- **Direction** shows where one place is compared to another.
- It helps us know how to move from one point to another.
- Example: *The library is north of the school.*

Types of Direction

1. **Egocentric Direction:**
   - ✓ Based on your position (e.g., left, right, front, back).
   - ✓ Example: *The park is behind me.*
2. **Landmark-Based Direction:**
   - ✓ Uses well-known places as references.
   - ✓ Example: *The bus stop is next to the supermarket.*

GEOGRAPHIC CONCEPTS

3. Geographic Directions

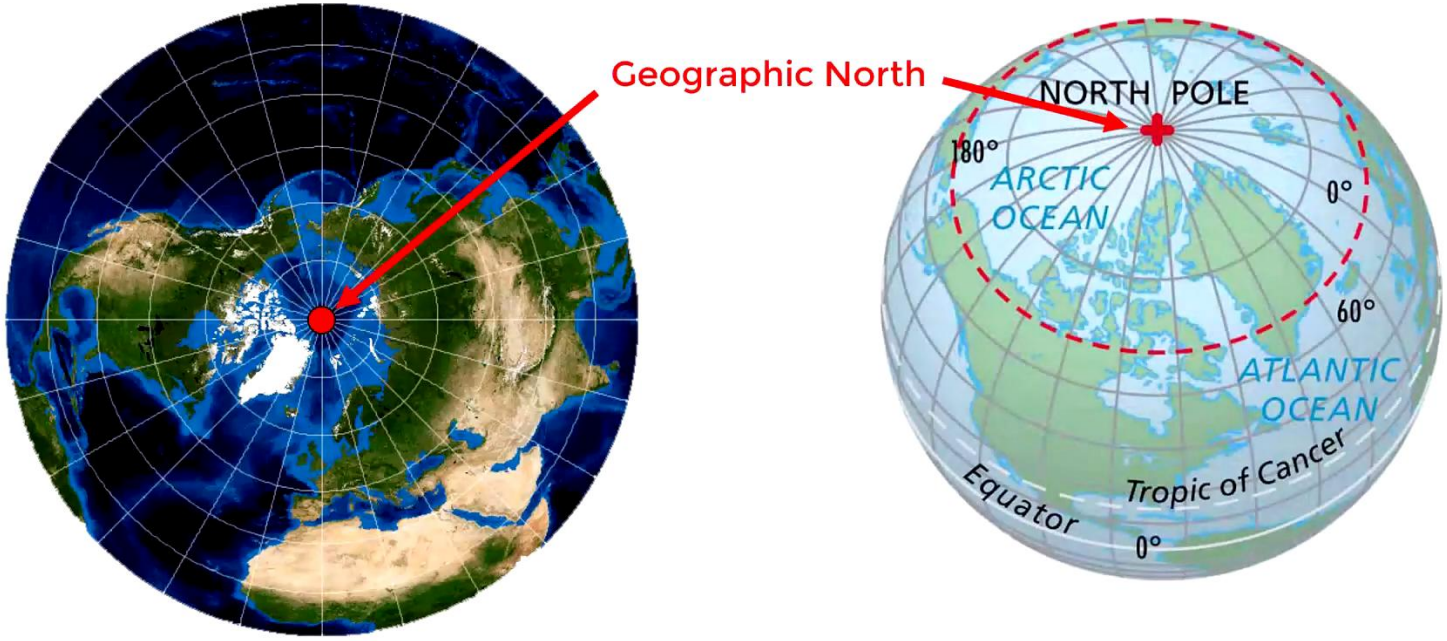
1. **True North:** Points to the North Pole.
2. **Magnetic North:** Points to Earth's magnetic field (used by compasses).
3. **Grid North:** Points along the grid lines on maps (latitude and longitude).

✓ In GIS, direction helps us create accurate maps and plan routes.

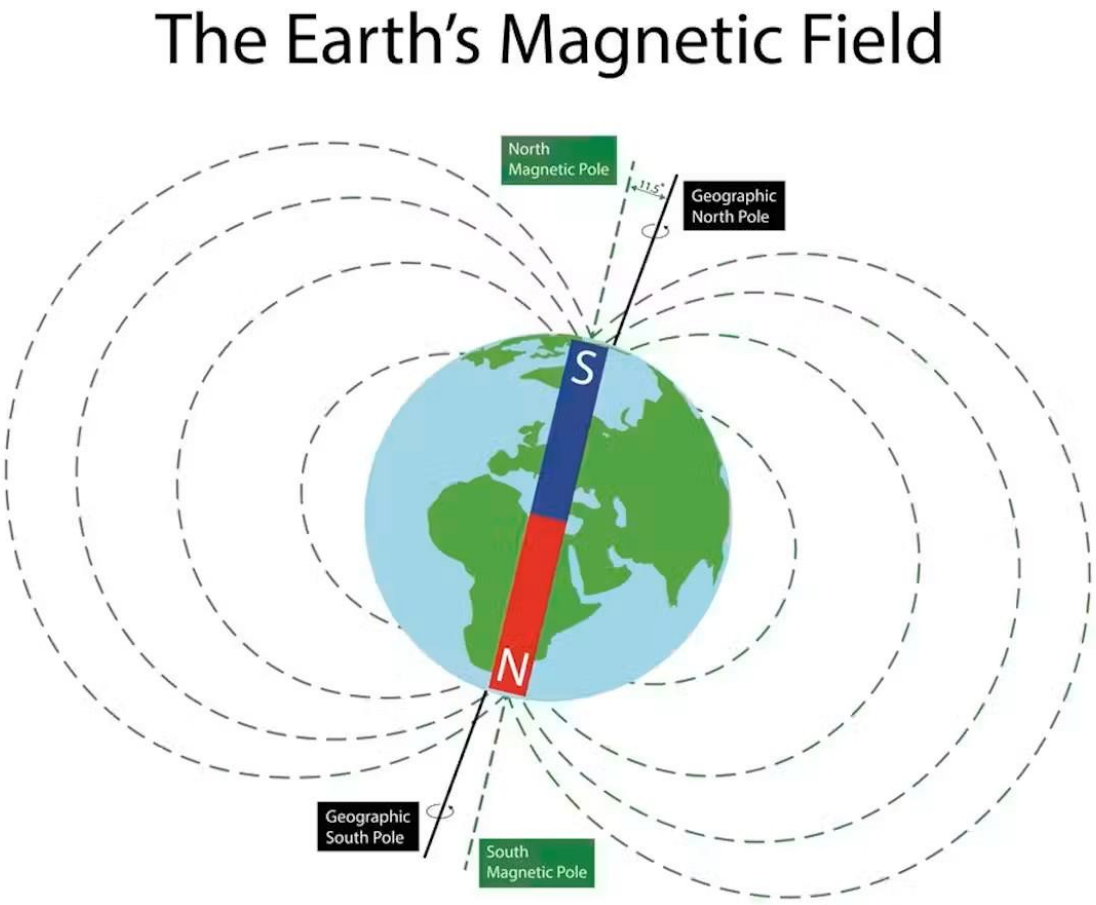
GEOGRAPHIC CONCEPTS

GEOGRAPHIC NORTH POLE ➡ True North

It is the "real" north pole of the earth, where the meridians converge and through which the planet's axis of rotation passes.



GEOGRAPHIC CONCEPTS



GEOGRAPHIC CONCEPTS

Distance

- Distance is an important concept in geography and GIS.
- It tells us how far one place is from another.
- Understanding distance helps us measure, compare, and analyze locations.

Types of Distance :

1. Nominal Distance (Descriptive Distance)
2. Absolute Distance (Measured Distance)
3. Relative Distance

GEOGRAPHIC CONCEPTS

➢ Nominal Distance (Descriptive Distance)

- Uses words to describe distance instead of numbers.
- Examples:
  - ✓ *The market is far from my house.*
  - ✓ *The school is near the park.*
- This type of distance is **subjective** because "near" or "far" may mean different things to different people.

GEOGRAPHIC CONCEPTS

➢ Absolute Distance (Measured Distance)

- Exact measurement between two points using standard units.
- Common units:
  - ✓ Meters, kilometers (km), miles (mi), feet.
- Examples:
  - ✓ *The distance from City A to City B is 150 km.*
  - ✓ *The river is 500 meters from the road.*

GEOGRAPHIC CONCEPTS

➢ Relative Distance

- Measures distance based on time, effort, or cost rather than actual physical length.
- Examples:
  - ✓ *Two cities may be 50 km apart, but if one has a fast highway and the other has a rough road, the travel time will be different.*
  - ✓ *Walking 1 km uphill is harder than walking 1 km on flat ground.*
  - ✓ *A flight from Paris to London is 1 hour, while a train takes 4 hours, even though the distance is the same.*
- This type of distance is **useful for navigation and planning** in GIS.

GEOGRAPHIC CONCEPTS

How GIS Uses Distance

- GIS helps measure and analyze distances for many applications, such as:
  1. **Urban Planning:** Finding the best locations for schools, hospitals, and parks based on population distance.
  2. **Transportation:** Creating shortest-route maps for cars, buses, and deliveries.
  3. **Disaster Management:** Measuring how far flood-prone areas are from safe zones.
  4. **Environmental Studies:** Analyzing distances between forests, rivers, and cities to study pollution effects.

GEOGRAPHIC CONCEPTS

- **Measuring Distance in GIS**

- GIS tools measure distances in two ways:

1. **Euclidean Distance (Straight-Line Distance):**
    1. The shortest path between two points (as the crow flies).
    2. Example: The distance between two cities measured directly on a map.
2. **Network Distance (Route-Based Distance):**
    1. Follows roads, rivers, or other pathways.
    2. Example: A car route that follows streets instead of a straight line.

GEOGRAPHIC CONCEPTS

Space

- **Space** is an area or surface on Earth where objects are located.
- In GIS, we study how objects like cities, rivers, and buildings are arranged in space.
- Example: Houses in a city can be close together or spread far apart.

Types of Space

1. **Physical Space:** Actual places on Earth (like mountains, rivers).
2. **Abstract Space:** Concepts like "public space" (like parks) or "private space" (like your home).
3. **Topological Space:** Focuses on how places are connected, not the actual distance.

For *example:* A subway map shows how stations are linked, even if the distance between them is different in real life.

GEOGRAPHIC CONCEPTS

Space

- **Why Space Matters in GIS ?**
- **Space helps GIS:**
  - ✓ Show where objects are located.
  - ✓ Analyze patterns (e.g., where most schools are in a city).
  - ✓ Understand relationships (e.g., how close a hospital is to homes).

(This metro map from Moscow is a topological space)



GEOGRAPHIC CONCEPTS

Navigation

- **What is Navigation?**
- ✓ **Navigation** is the process of moving from one place to another.
- ✓ It helps us **find locations**, **plan routes**, and **avoid obstacles**.
- ✓ GIS plays a big role in modern navigation by providing **maps, directions, and travel data**.

GEOGRAPHIC CONCEPTS

Types of Navigation

1. Land Navigation
2. Marine Navigation
3. Air Navigation
4. Space Navigation

GEOGRAPHIC CONCEPTS

Land Navigation

- Movement across the land using maps, signs, or landmarks.
- **Examples:**
  - ✓ Walking to a store using street signs.
  - ✓ A car using GPS for turn-by-turn directions.

Marine Navigation

- Traveling over water using stars, maps, or GPS.
- **Examples:**
  - ✓ Sailors using a compass and nautical charts.
  - ✓ A ship using GPS to avoid shallow areas.

GEOGRAPHIC CONCEPTS

Air Navigation

- Pilots use instruments, weather data, and GPS to travel.
- **Examples:**
  - ✓ Airplanes following flight paths.
  - ✓ Drones using GPS for automated flight.

Space Navigation

- Used for satellites and spacecraft.
- **Example:**
  - ✓ NASA using star positions to guide spacecraft.

GEOGRAPHIC CONCEPTS

Three Types of Geographic Knowledge in Navigation

1. **Landmark Knowledge:**
    1. Recognizing unique features like buildings or mountains.
    2. Example: *Knowing that your school is next to a big park.*
2. **Route Knowledge:**
    1. Understanding **which roads or paths** connect different places.
    2. Example: *Knowing the way from home to school using main streets.*
3. **Survey Knowledge:**
    1. Understanding the **overall layout** of an area.
    2. Example: *Knowing where different neighborhoods are located in a city.*

GEOGRAPHIC CONCEPTS

- **Navigation Tools & Technologies**
- **Traditional Tools:**
  - Maps, compass, stars.
- **Modern Tools:**
  - GPS (Global Positioning System).
  - Mobile apps (Google Maps, Waze).
  - GIS-based navigation systems.

GEOGRAPHIC CONCEPTS

Challenges in Navigation

1. **Changing Maps:** New roads and buildings require frequent updates.
2. **Traffic & Road Conditions:** Accidents and closures affect travel time.
3. **GPS Errors:** Sometimes signals are weak or inaccurate.
4. **Weather Conditions:** Fog, storms, and snow can make navigation difficult.

Chapter 2

Map Anatomy

MAP ANATOMY



MAP ANATOMY

What is a Map?

- A map is a visual representation of an area that shows locations, features, and relationships between places.
- Maps can be used to navigate, analyze data, and make decisions.
- Maps simplify reality by showing only important information.

Key Characteristics of Maps:

- ✓ Represent locations using symbols, colors, and labels.
- ✓ Can be static (printed or digital) or dynamic (digital & interactive).
- ✓ Show spatial relationships between features.

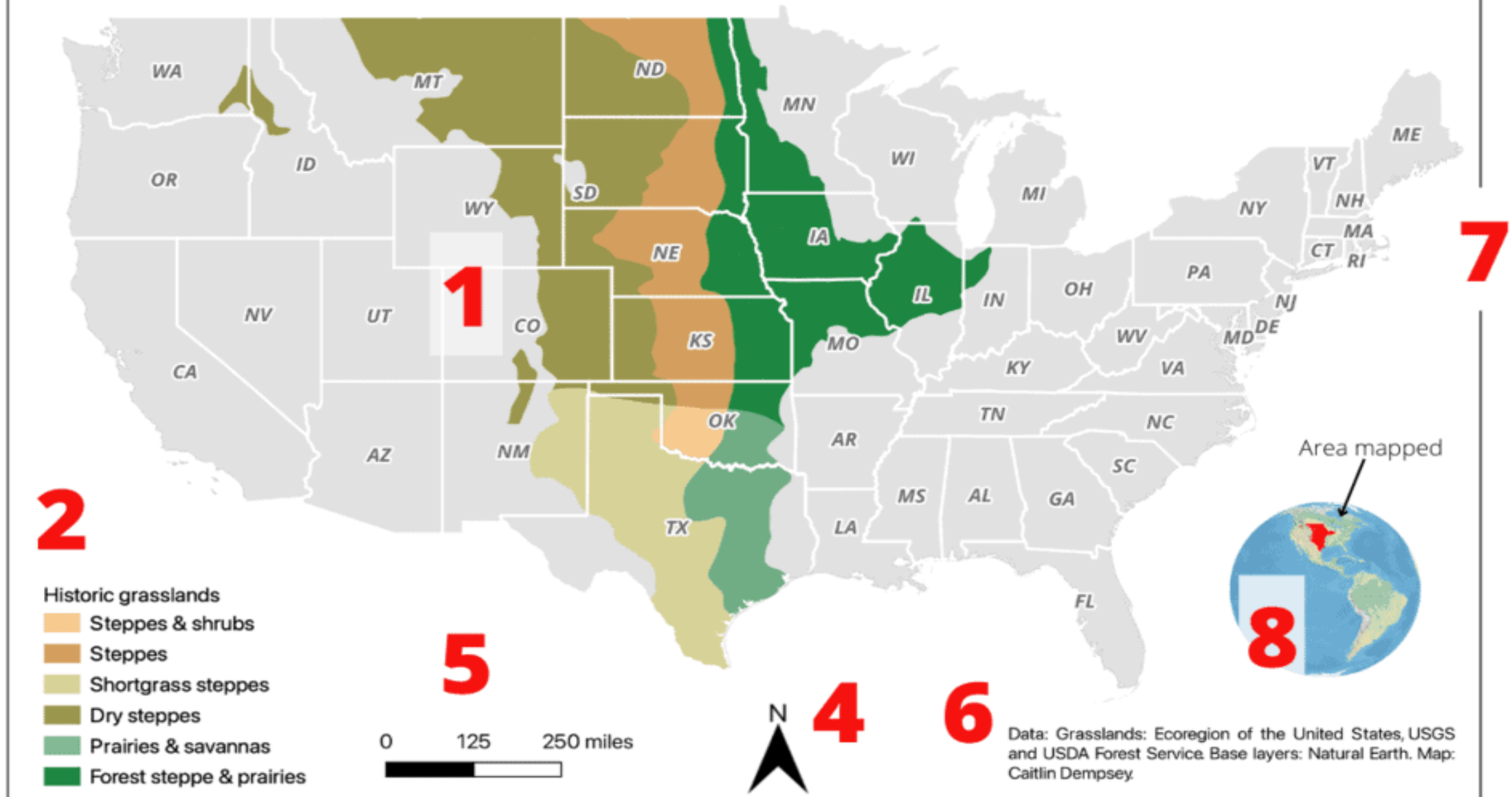
COMPONENTS OF A MAP

- Every map contains important elements that help users understand its information.

Main Map Components:

1. **Title:** Describes what the map represents.
2. **Legend (Key):** Explains symbols and colors used.
3. **Scale:** Shows real-world distances on the map.
4. **Compass (Direction Indicator):** Shows cardinal directions (North, South, East, West).
5. **Grid & Coordinates:** Helps locate places accurately.

Historic grasslands in the continental United States



Data: Grasslands: Ecoregion of the United States, USGS and USDA Forest Service. Base layers: Natural Earth. Map: Caitlin Dempsey.

COMPONENTS OF A MAP

Map components for previous slide:

note: some components are optional while some other components are really important and without them our map will be meaningless.

1. data (or map) frame
2. map legend
3. map title
4. north arrow
5. map scale bar
6. metadata (or map citation)
7. border
8. inset (or locator) map.

CATEGORIES OF MAPS

Categories of Maps

- Maps are divided into three major categories:
  1. **Reference Maps** – Show geographic locations of places and features.
  2. **Thematic Maps** – Focus on a specific theme, such as population or climate.
  3. **Dynamic Maps** – Interactive maps that update in real-time.
- Each type of map serves a different purpose and is used in GIS for different applications.

CATEGORIES OF MAPS

Reference Maps

- Reference maps are used to locate physical and human-made features accurately. They provide a general overview of an area.
- Common Features of Reference Maps:
  - ✓ Show boundaries, roads, cities, rivers, and other landmarks.
  - ✓ Often used for navigation and planning.
  - ✓ Represent features accurately and consistently.

Please search on the internet and find some reference maps

CATEGORIES OF MAPS

Types of Reference Maps

1- Political Maps

- ✓ Show country, state, and city boundaries.
- ✓ Help with government planning, elections, and administration.
- ✓ Example: A world map showing country borders.

2- Physical Maps

- ✓ Display natural features like mountains, rivers, and lakes.
- ✓ Use shading and colors to represent elevation.
- ✓ Example: A map of Europe showing mountain ranges and rivers.

3- Road Maps

- ✓ Focus on streets, highways, and transportation routes.
- ✓ Used for driving, walking, and biking navigation.
- Example: Google Maps showing driving directions.

4- Topographic Maps

CATEGORIES OF MAPS

Topographic Maps (A Special Type of Reference Map)

- Topographic maps show elevation and terrain features.
- Use contour lines to represent changes in elevation.
- Applications:
  - ✓ Used by hikers, geologists, and military planners.
  - ✓ Helps engineers plan roads and construction projects.
- Example: A topographic map of a national park.

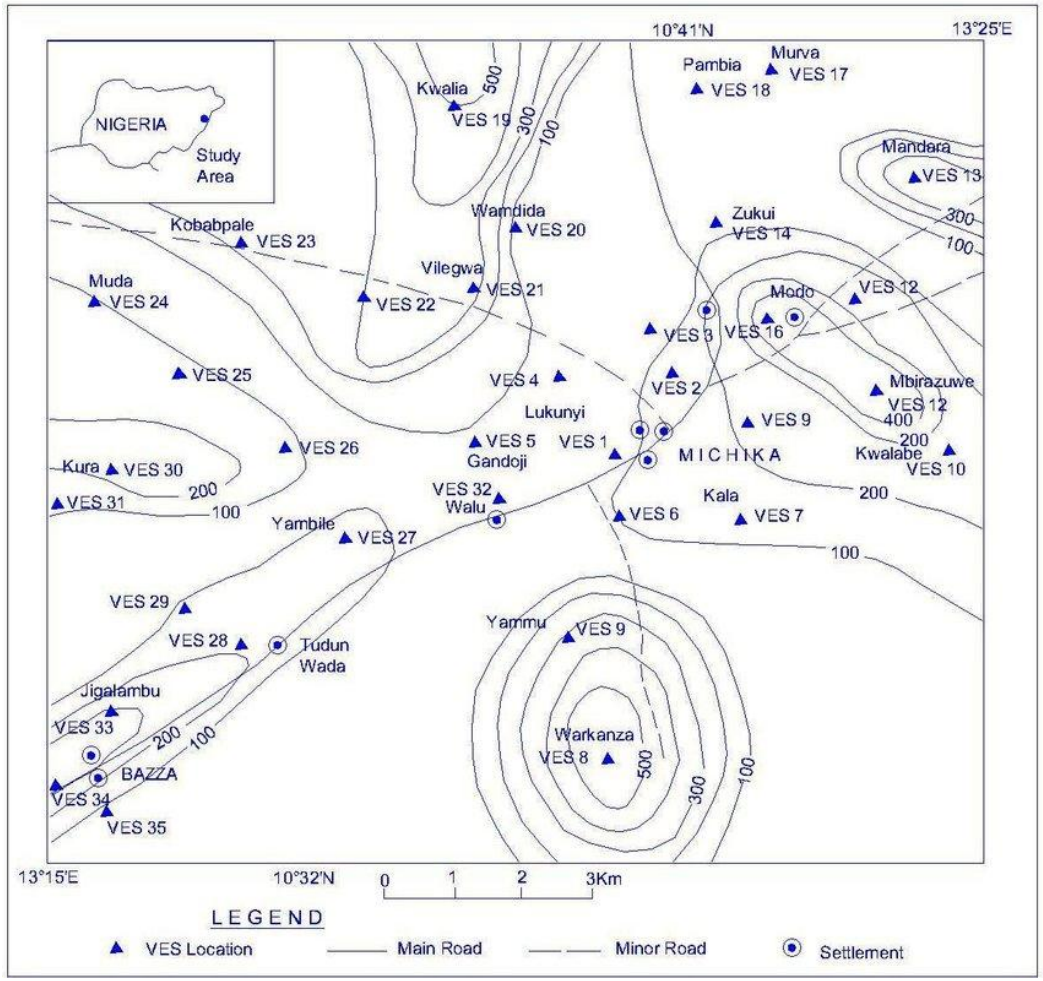
CATEGORIES OF MAPS



CATEGORIES OF MAPS



CATEGORIES OF MAPS



CATEGORIES OF MAPS

➢ Thematic Maps

- ❖ Thematic maps show specific information about an area rather than just locations.
- ❖ They help visualize patterns and trends.
- ❖ Common Features of Thematic Maps :
  - ✓ Focus on one subject (e.g., population, temperature, pollution).
  - ✓ Use colors, symbols, or shading to represent data.
  - ✓ Useful for analysis and decision-making in GIS.

CATEGORIES OF MAPS

Types of Thematic Maps

1- Choropleth Maps

- Use color shades to show differences in data values.
- Example: A map showing income levels in different regions.

2- Dot Density Maps

- Use dots to represent the number of features in an area.
- Example: A map showing earthquake locations with dots.

3- Proportional Symbol Maps

- Use different-sized symbols (e.g., circles) to represent data values.
- Example: A map where larger circles show cities with bigger populations.

4- Isoline (Contour) Maps

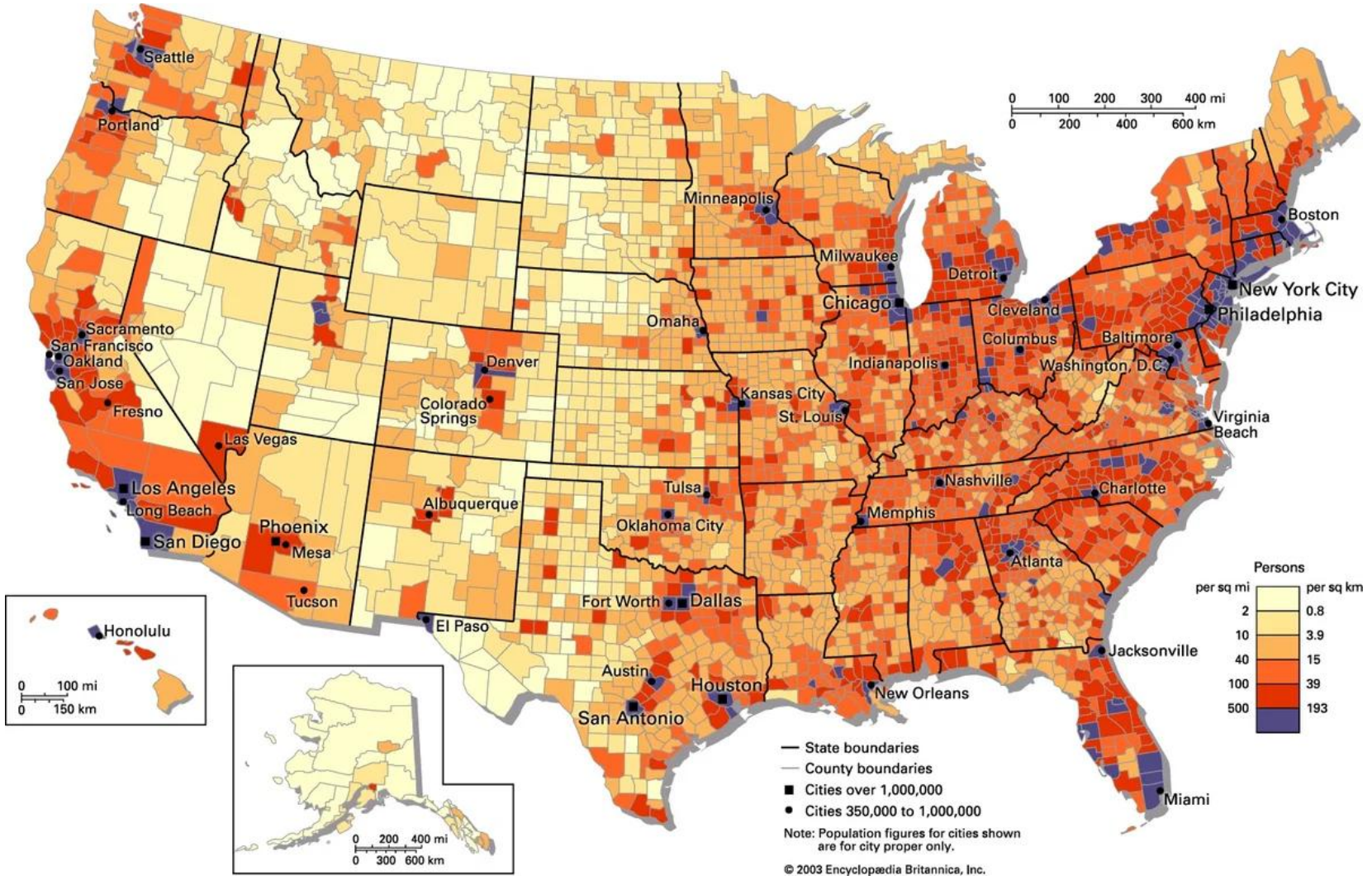
- Use lines to connect areas of equal value (e.g., elevation, temperature).
- Example: A weather map showing temperature zones.

5- Cartograms

- Distort the size of geographic areas based on data instead of real-world size.
- Example: A map where countries appear larger based on GDP

CATEGORIES OF MAPS

Most common type of thematic map is Choropleth Maps

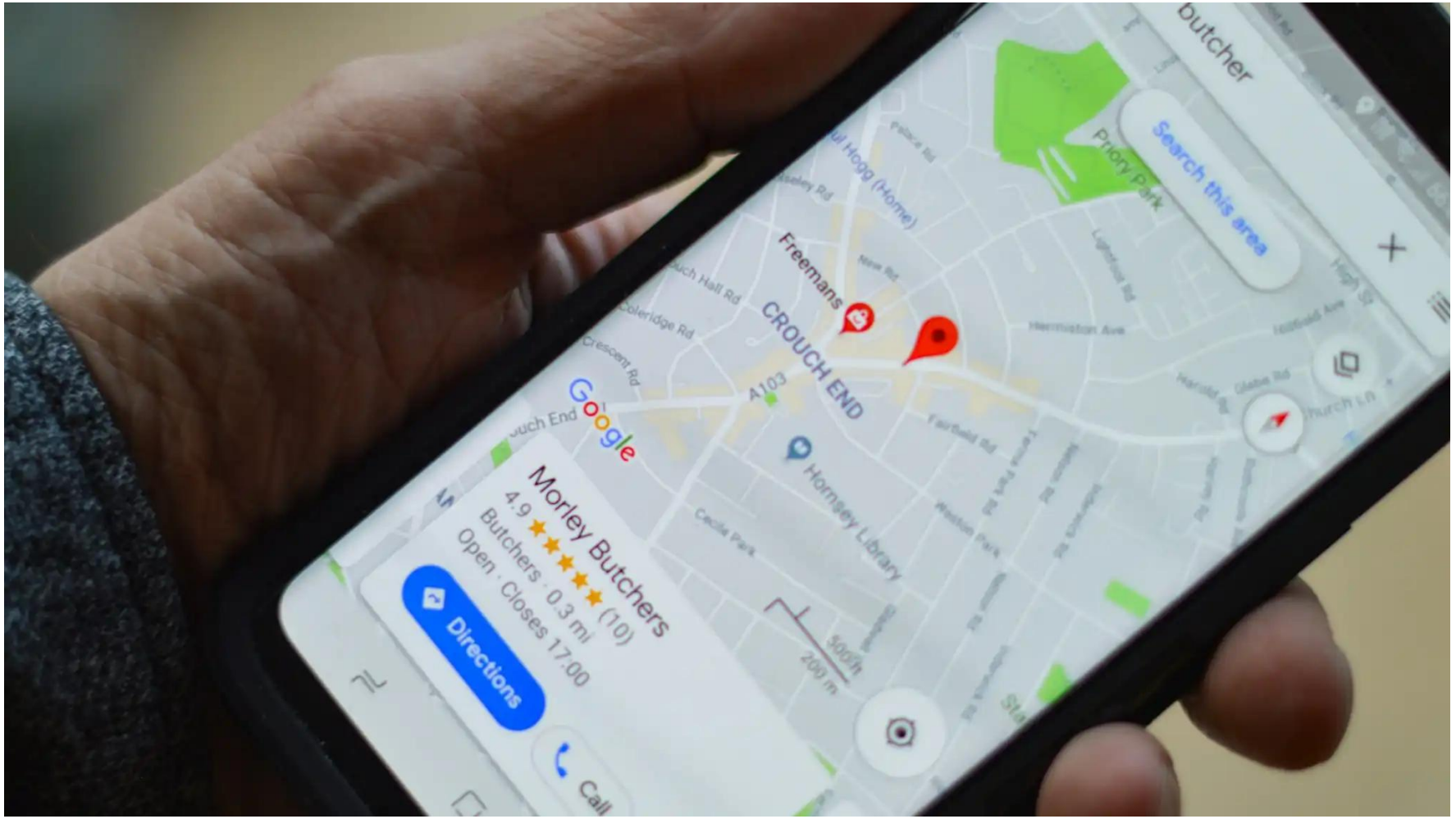


CATEGORIES OF MAPS

Dynamic Maps

- Dynamic maps are interactive, digital maps that update in real-time.
- Features of Dynamic Maps:
  - ✓ Users can zoom, pan, and select layers.
  - ✓ Show real-time changes (e.g., traffic, weather).
  - ✓ Found in GPS apps, social media, and online platforms.
- Examples:
  - ✓ Google Maps.
  - ✓ Online weather maps.

CATEGORIES OF MAPS



CATEGORIES OF MAPS

- GIS combines different types of maps to analyze and solve real-world problems.
- ✓ Reference Maps help us find locations.
- ✓ Thematic Maps help us study patterns.
- ✓ Dynamic Maps help us track changes over time.

MAP SCALE

Understanding the Relationship Between Maps and the Real World

What is Map Scale?

- Map scale is the ratio between distances on a map and real-world distances.
- It shows how much the real world has been reduced to fit on a map.
- Example: A scale of 1:100,000 means that 1 cm on the map equals 100,000 cm (1 km) in real life.

MAP SCALE

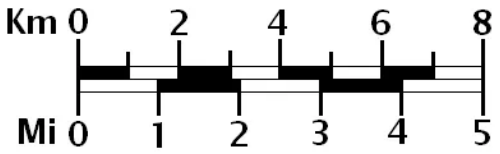
Why is Map Scale Important?

- Map scale affects:
  - ✓ How much detail is shown (more detail at large scales, less at small scales).
  - ✓ How measurements are taken (distances, areas).
  - ✓ How maps are used (for navigation, city planning, global studies).
- Example: A city street map needs a large scale to show individual buildings, but a world map needs a small scale to fit everything.

MAP SCALE

Three Ways to Represent Map Scale

1. **Verbal Scale (Word Statement)**
   - ✓ Written description of scale.
   - ✓ Example: "*One inch represents one mile.*"
2. **Graphic Scale (Scale Bar)**
   - ✓ A visual ruler on the map that helps measure distances.
   - ✓ Example:



3. **Representative Fraction (RF) or Ratio Scale**
   - ✓ Expressed as a fraction or ratio (e.g., **1:50,000**).
   - ✓ It is independent of the unit which means Works with **any measurement system** (cm, inches, meters).

MAP SCALE

➢ Understanding Representative Fraction (RF) Scale

- ❖ 1:10,000 → Large-scale map (shows a small area with more detail).
- ❖ 1:1,000,000 → Small-scale map (shows a large area with less detail).
- ❖ Example:
  - ✓ A 1:25,000 map shows a small city with street details.
  - ✓ A 1:5,000,000 map shows a whole country with just major cities.

MAP SCALE


Map1
NEW YORK
CUNY-Hunter College
Queensboro Brg Lo...
9A
495
1 Inch = 1 Mile (1:63,360)
0
.5
1
Miles
Maptitude
www.MappingSoftware.com

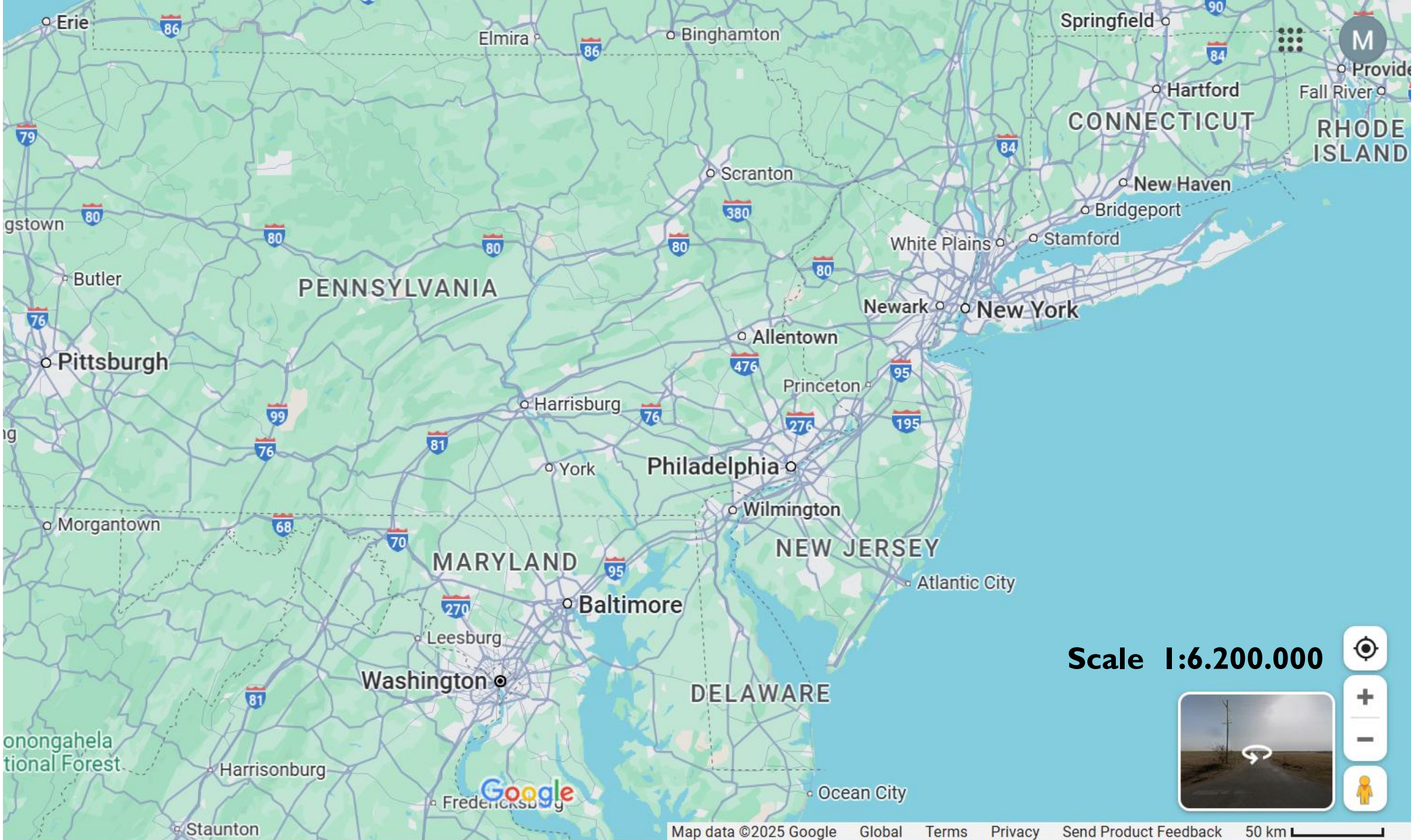
MAP SCALE

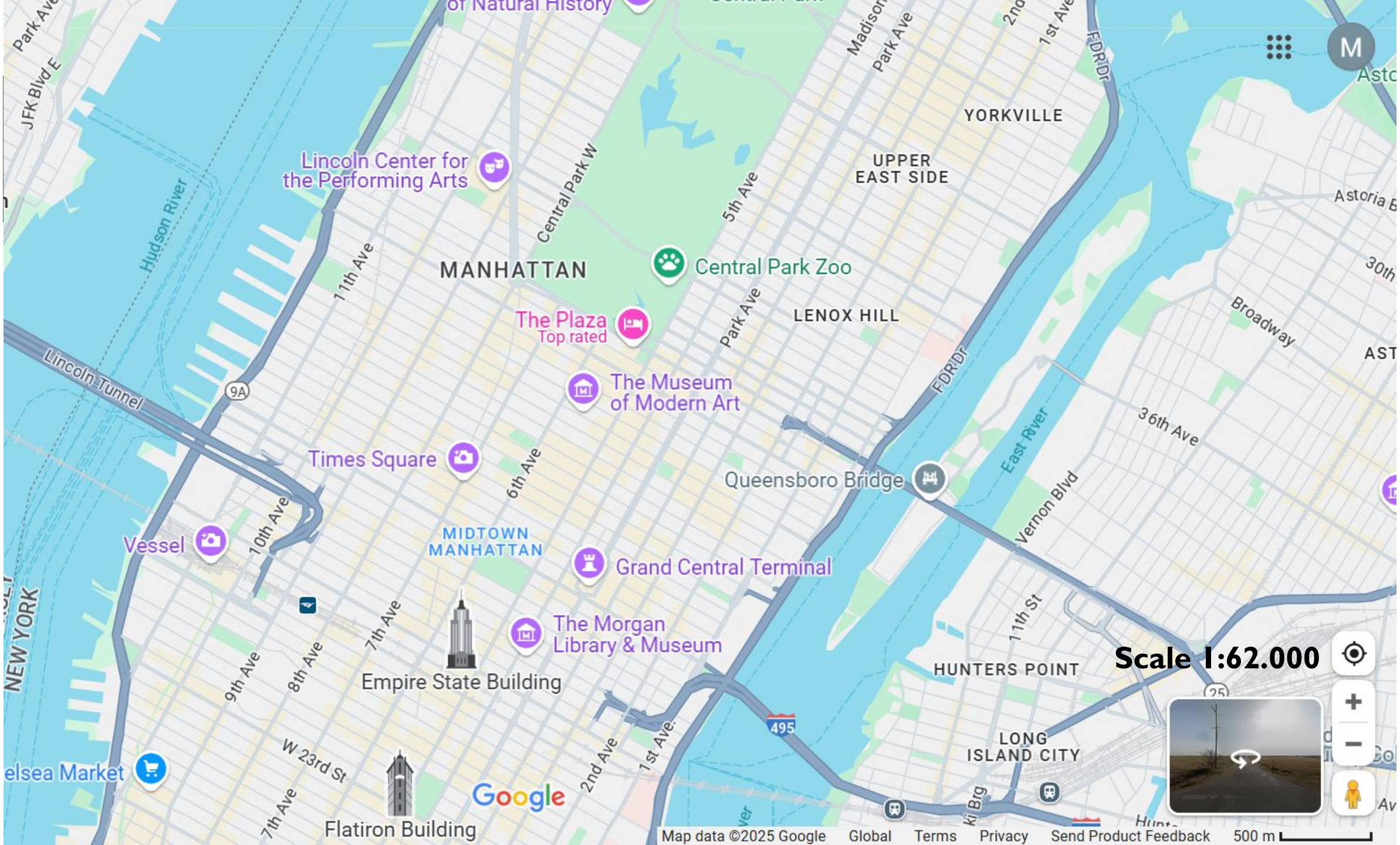
Find a map with ratio scale on the internet

MAP SCALE

➢ Understanding Representative Fraction (RF) Scale

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Erie
Elmira
Binghamton
Springfield
Hartford
CONNECTICUT
RHODE ISLAND
Fall River
Scranton
New Haven
Bridgeport
White Plains
Stamford
Butler
PENNSYLVANIA
Newark
New York
Allentown
Pittsburgh
Princeton
Harrisburg
Philadelphia
York
Wilmington
Morgantown
NEW JERSEY
MARYLAND
Atlantic City
Baltimore
Leesburg
Washington
DELAWARE
Harrisonburg
Ocean City
Staunton
Google
Scale 1:6.200.000
Map data ©2025 Google
Global
Terms
Privacy
Send Product Feedback
50 km


Lincoln Center for the Performing Arts
Central Park W
5th Ave
UPPER EAST SIDE
YORKVILLE
Hudson River
11th Ave
MANHATTAN
Central Park Zoo
The Plaza
Top rated
Park Ave
LENOX HILL
Broadway
Lincoln Tunnel
9A
The Museum of Modern Art
FDR Dr
Times Square
6th Ave
Queensboro Bridge
East River
36th Ave
Vessel
10th Ave
MIDTOWN MANHATTAN
Grand Central Terminal
Vernon Blvd
NEW YORK
7th Ave
8th Ave
9th Ave
The Morgan Library & Museum
Empire State Building
11th St
HUNTERS POINT
Scale 1:62.000
495
LONG ISLAND CITY
W 23rd St
2nd Ave
1st Ave
elsea Market
Google
Flatiron Building
Map data ©2025 Google
Global
Terms
Privacy
Send Product Feedback
500 m

MAP SCALE

Large-Scale vs. Small-Scale Maps

- Large-Scale Maps (More Detail, Less Area Covered)
  - ✓ Cover small areas (e.g., neighborhoods, towns).
  - ✓ Show detailed features (roads, buildings, parks).
  - ✓ Example: 1:10,000 map of a city.
- Small-Scale Maps (Less Detail, More Area Covered)
  - ✓ Cover large areas (e.g., countries, States).
  - ✓ Show less detail, only major features.
  - ✓ Example: 1:5,000,000 map of a country.

MAP SCALE

- **Some examples for choosing the right scale for different uses**

- ✓ 1:1,000 – 1:10,000: Used for building plans, urban mapping.
- ✓ 1:50,000 – 1:100,000: Used for topographic maps, hiking maps.
- ✓ 1:250,000 – 1:1,000,000: Used for regional planning, military maps.
- ✓ 1:5,000,000 – 1:50,000,000: Used for world maps, airline routes.

MAP SCALE

Question : Which scale is best for **finding a restaurant** vs. **planning a flight ?**

MAP SCALE

GIS and Map Scale

- GIS can adjust scale dynamically (zoom in/out).
- Different GIS layers have different scales:
  - ✓ Satellite images (large-scale for details).
  - ✓ Political maps (small-scale for big areas).
- GIS allows interactive scaling, meaning:
  - ✓ Users zoom in for details.
  - ✓ Users zoom out for general patterns.

COORDINATE SYSTEMS

What is a Coordinate System?

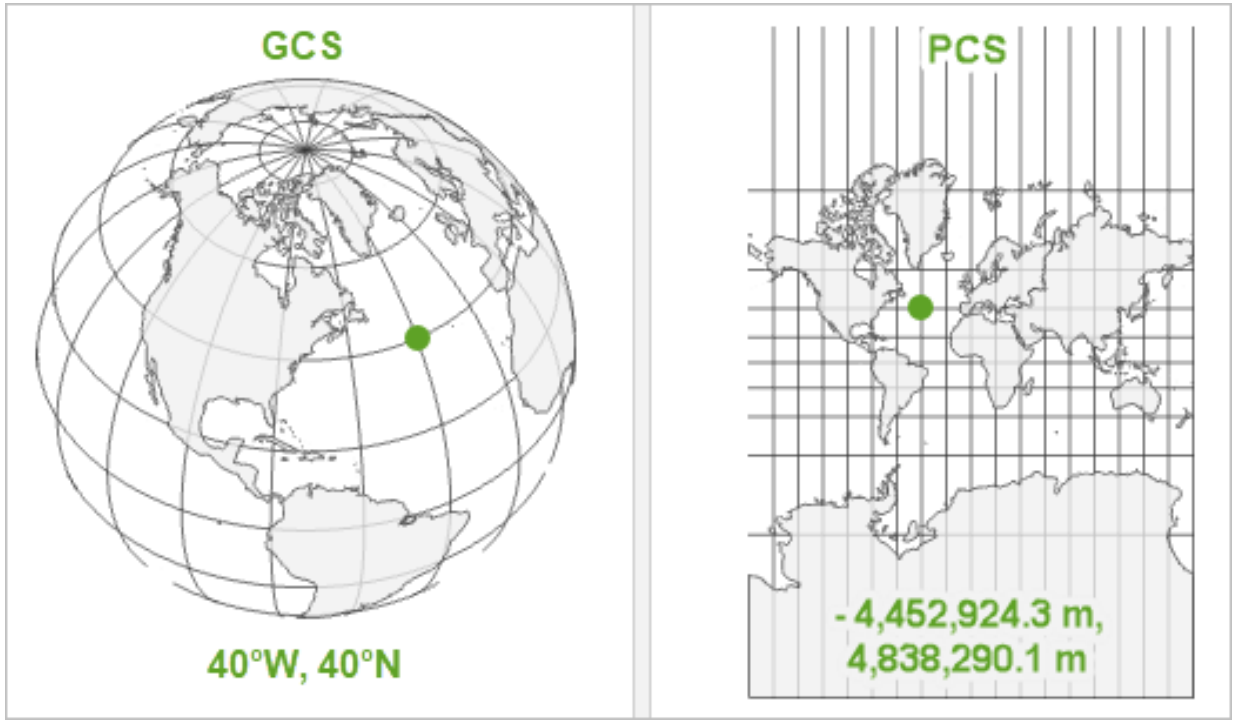
- A coordinate system is a framework used to define exact positions on Earth's surface using numbers.
- It allows us to locate, map, and navigate places accurately.
- Example: GPS uses coordinate systems to give locations.
- **Why Do We Need Coordinate Systems?**
  - ✓ The Earth is large and round, so we need a system to describe locations accurately.
  - ✓ Without a coordinate system, finding locations would be confusing.

COORDINATE SYSTEMS

➢ Types of Coordinate Systems

❖ There are **two main types** of coordinate systems that each system has different uses and advantages.

1. **Geographic Coordinate System (GCS)** – Based on latitude and longitude.
2. **Projected Coordinate System (PCS)** – Converts Earth's curved surface into a flat map.



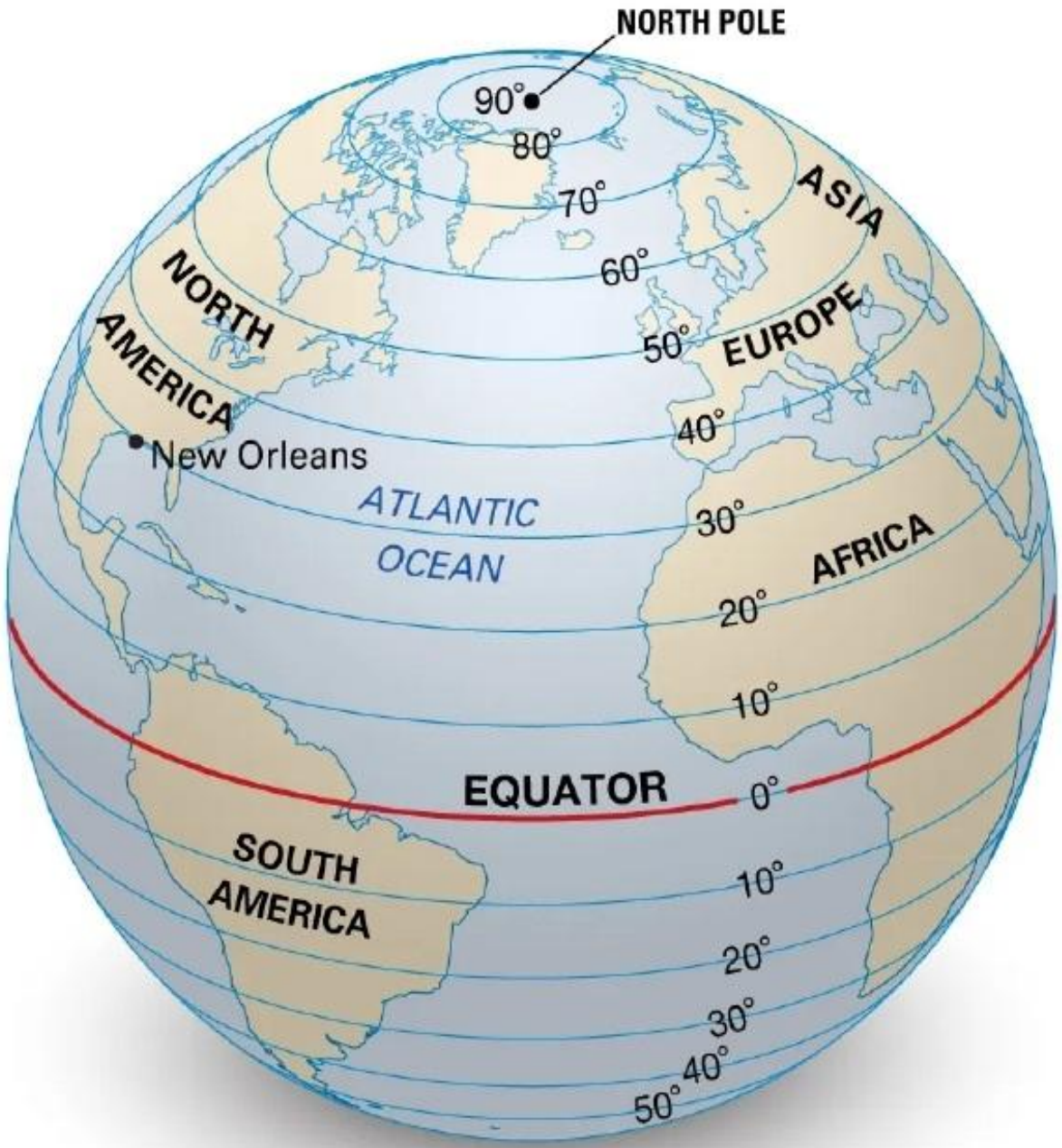
GEOGRAPHIC COORDINATE SYSTEM (GCS)

➢ What is the Geographic Coordinate System (GCS)?

- ✓ The most common way to locate points on Earth.
- ✓ Uses a spherical (3D) model of the Earth.
- ✓ It's based on **latitude** and **longitude**.
- ✓ Used in GPS, global mapping, and world navigation.

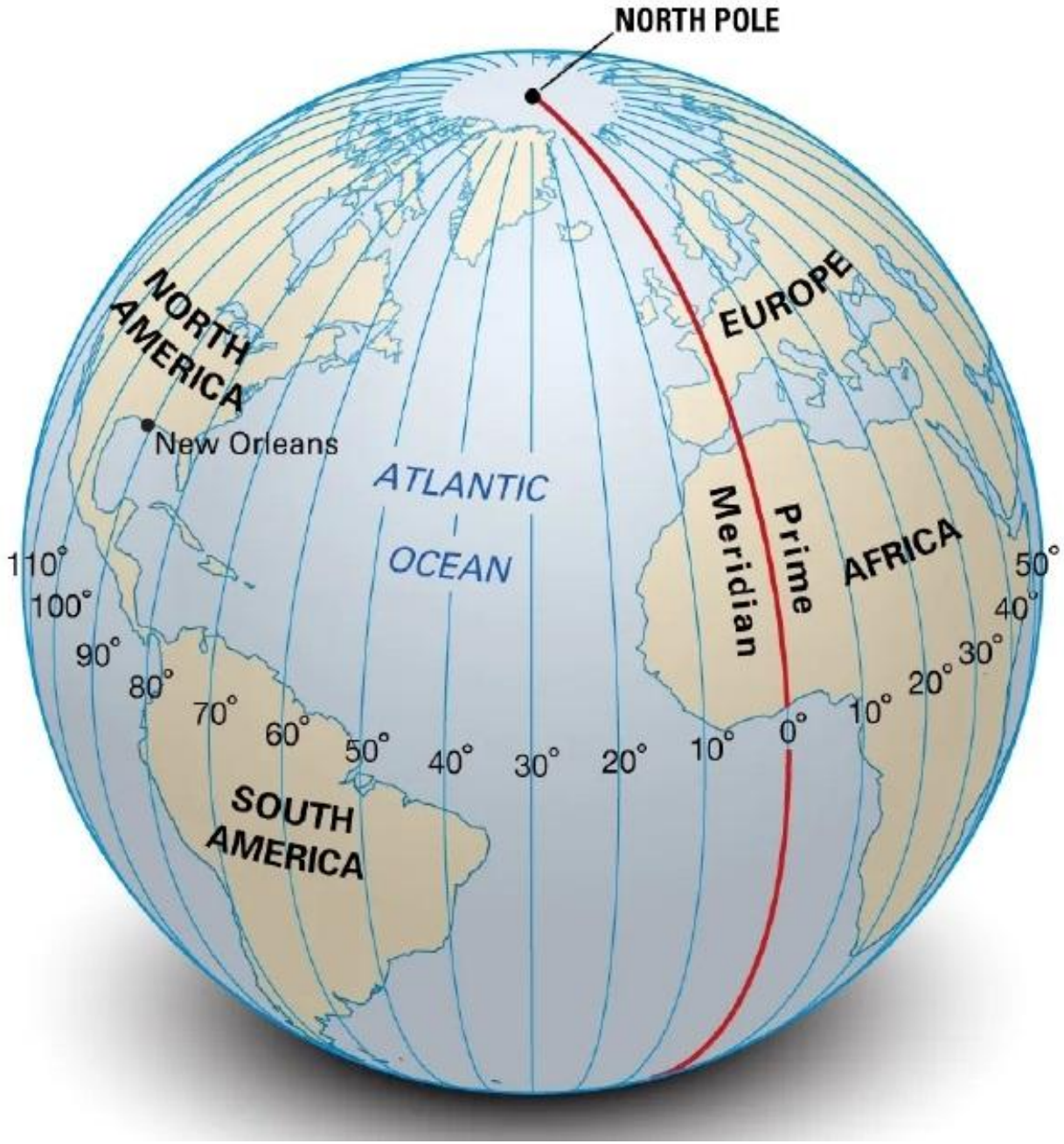
GEOGRAPHIC COORDINATE SYSTEM (GCS)





FACTS ABOUT LINES OF LATITUDE

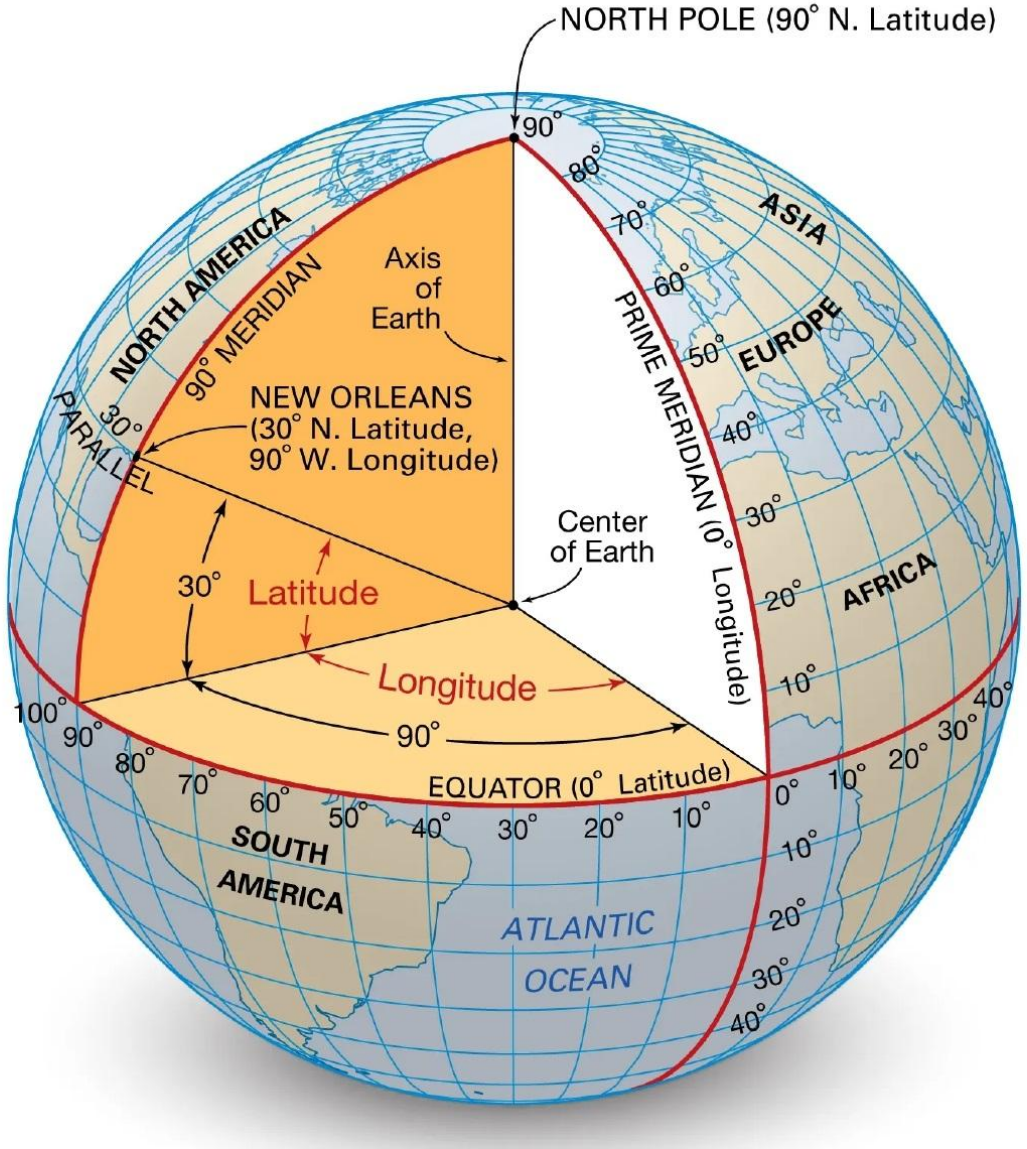
- Are known as parallels.
- Run in an east-west direction.
- Measure distance north or south from the Equator.
- Are parallel to one another and never meet.
- Cross the prime meridian at right angles.
- Lie in planes that cross the Earth's axis at right angles.
- Get shorter toward the poles, with only the Equator, the longest, a great circle.

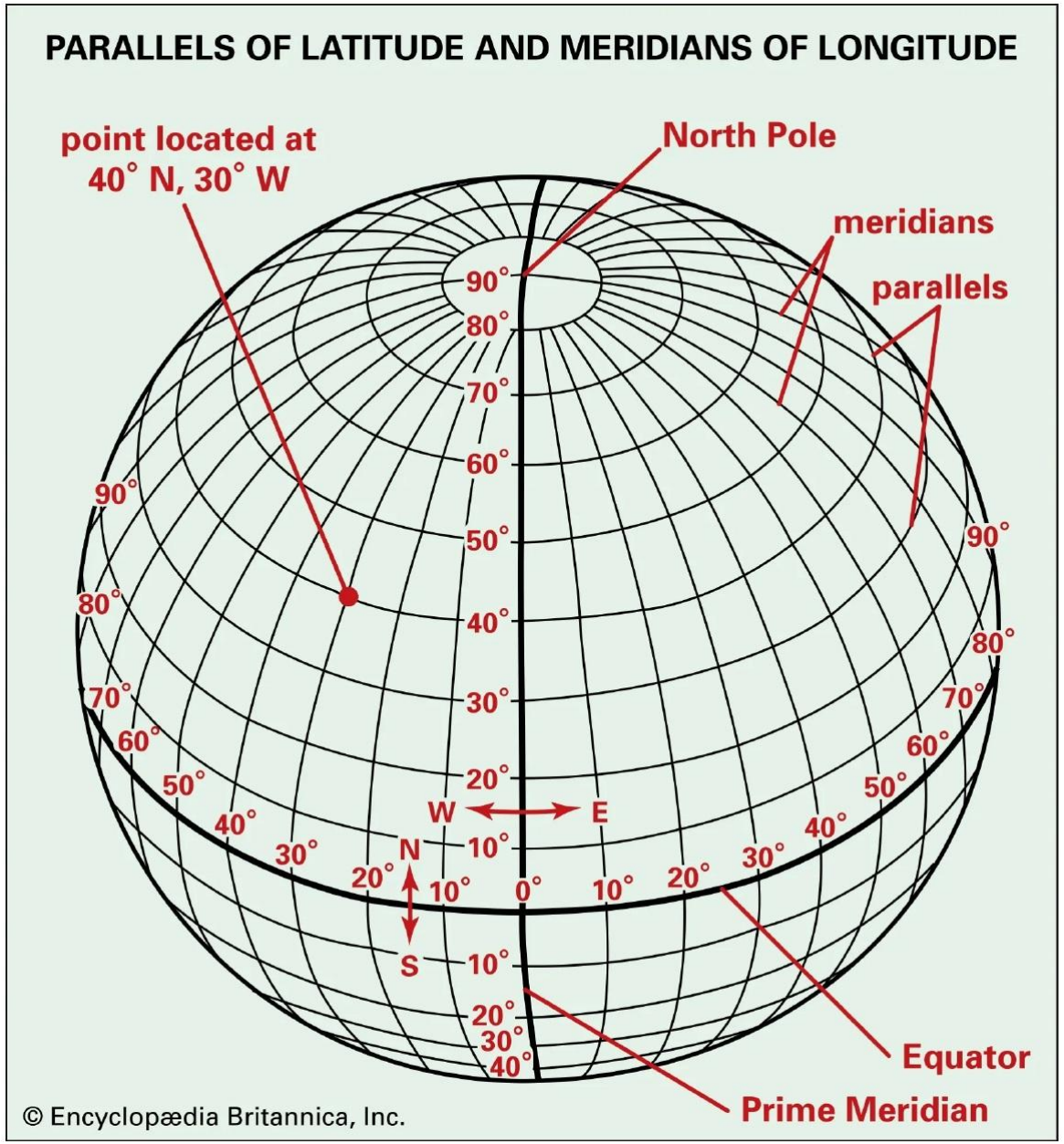


FACTS ABOUT LINES OF LONGITUDE

- Are known as meridians.
- Run in a north-south direction.
- Measure distance east or west of the prime meridian.
- Are farthest apart at the Equator and meet at the poles.
- Cross the Equator at right angles.
- Lie in planes that pass through the Earth's axis.
- Are equal in length.
- Are halves of great circles.


ASIA
NORTH POLE
90°
PRIME MERIDIAN
60°
EUROPE
39° PARALLEL
30°
NORTH AMERICA
AFRICA
PACIFIC OCEAN
WASHINGTON, D.C.
39° North Latitude
77° West Longitude
0°
ATLANTIC OCEAN
150°
EQUATOR
120°
30°
90°
60°
77° MERIDIAN
SOUTH AMERICA


NORTH POLE (90° N. Latitude)
90°
80°
70°
60°
50°
40°
30°
20°
10°
0°
NORTH AMERICA
90° MERIDIAN
Axis of Earth
NEW ORLEANS
(30° N. Latitude,
90° W. Longitude)
30° PARALLEL
ASIA
EUROPE
PRIME MERIDIAN (0° Longitude)
AFRICA
Center of Earth
30°
Latitude
Longitude
90°
100°
90°
80°
70°
60°
50°
40°
30°
20°
10°
EQUATOR (0° Latitude)
SOUTH AMERICA
ATLANTIC OCEAN


PARALLELS OF LATITUDE AND MERIDIANS OF LONGITUDE
point located at
40° N, 30° W
North Pole
meridians
parallels
90°
80°
70°
60°
50°
40°
30°
20°
10°
0°
W
E
N
S
Equator
Prime Meridian
© Encyclopædia Britannica, Inc.

GEOGRAPHIC COORDINATE SYSTEM (GCS)

➢ Understanding Latitude

- ❖ Latitude lines run east to west, but measure north to south.
- ❖ Equator (0° latitude) is the starting point.
- ❖ Latitude values:
  - ✓ North of the Equator: 0° to 90°N
  - ✓ South of the Equator: 0° to 90°S

✅ Example: London is at 51°N latitude.

➢ Understanding Longitude

- ❖ Longitude lines run north to south, but measure east to west.
- ❖ Prime Meridian (0° longitude) is the starting point.
- ❖ Longitude values:
  - ✓ East of Prime Meridian: 0° to 180°E
  - ✓ West of Prime Meridian: 0° to 180°W

✅ Example: New York is at 74°W longitude.

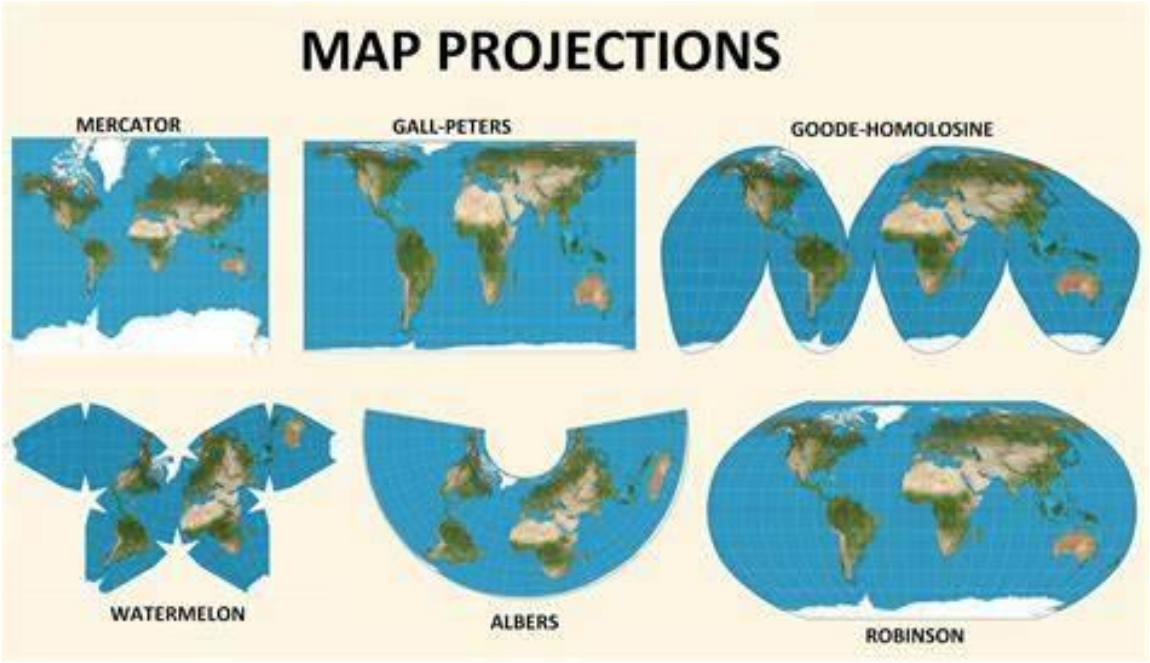
GEOGRAPHIC COORDINATE SYSTEM (GCS)

Latitude and Longitude Together

- Combining latitude and longitude gives an exact location.
- Example:
  - ✓ Paris, France: 48.8566° N, 2.3522° E
  - ✓ Sydney, Australia: 33.8688° S, 151.2093° E
- Every place on Earth has a unique coordinate.

PROJECTED COORDINATE SYSTEM (PCS)

What is the Projected Coordinate System (PCS)?



- ✓ Since the Earth is a 3D sphere, we can't show it perfectly on a flat map or screen.
- ✓ PCS transforms 3D locations into a 2D map.
- ✓ Projections help us transform the Earth's curved surface into a flat map, but some distortions (like in size or shape) will occur.
- 📌 Example: City planning requires flat maps, not curved ones.

PROJECTED COORDINATE SYSTEM (PCS)

➢ Common Projection Types in PCS

- Different projections are used for different regions and purposes.
- Some common projections:

1. **Mercator Projection**
2. **Universal Transverse Mercator (UTM)**

PROJECTED COORDINATE SYSTEM (PCS)

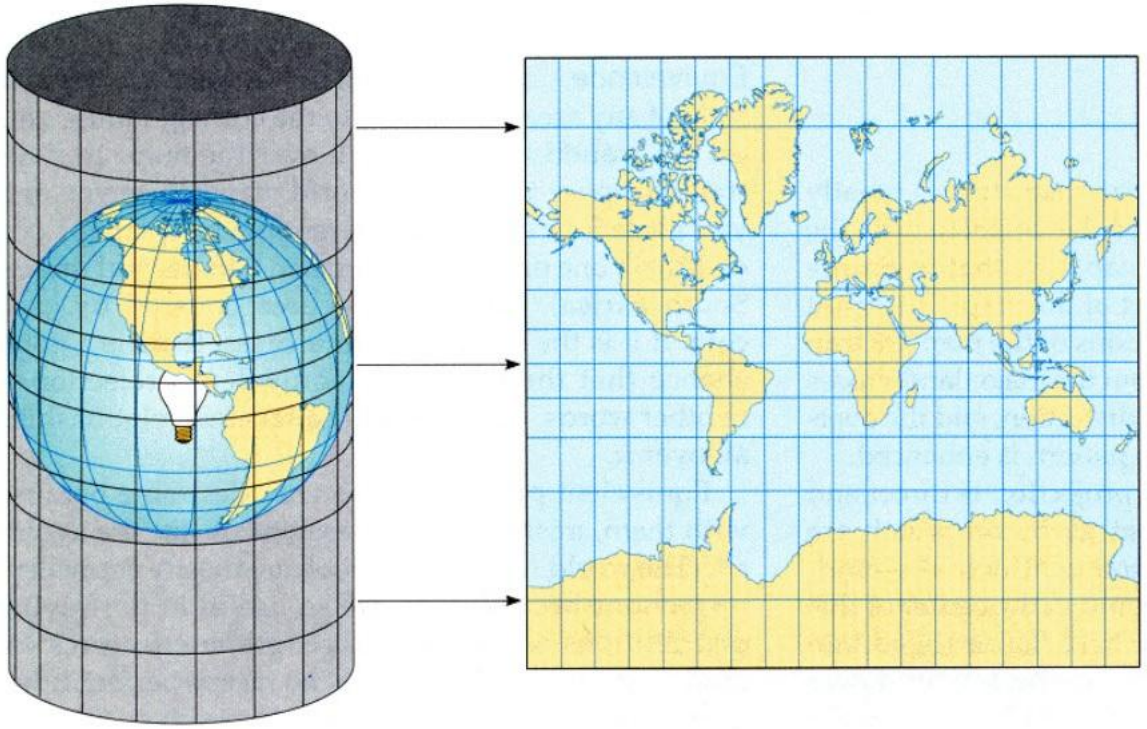
How Does the Mercator Projection Work?

- ✓ The Earth is a sphere, but the Mercator projection stretches it into a rectangle.
- ✓ Steps in the projection process:

1. Imagine wrapping a cylinder around the Earth, touching it at the Equator.
2. Project every point from the globe onto the cylinder.
3. Unroll the cylinder into a flat map.

- ✓ The Equator remains unchanged, but areas farther from it stretch larger and larger.

📌 Example: Greenland appears as big as Africa, but in reality, Africa is 14 times larger!



PROJECTED COORDINATE SYSTEM (PCS)

Let's watch a video about map projection and

Mercator map projection system

PROJECTED COORDINATE SYSTEM (PCS)

➢ Where is the Mercator Projection Used Today?

📌 Navigation & GPS Maps:

✓ Used in Google Maps, Bing Maps, and OpenStreetMap because it preserves direction.

✓ Airplane pilots use it for route planning.

📌 Web-Based Mapping Applications:

✓ Many interactive maps (like Google Earth) use Mercator projection.

📌 Historical and Nautical Maps:

✓ Still used in marine charts and navigation systems.

PROJECTED COORDINATE SYSTEM (PCS)

How Does UTM Work?

📌 Key Features of UTM:

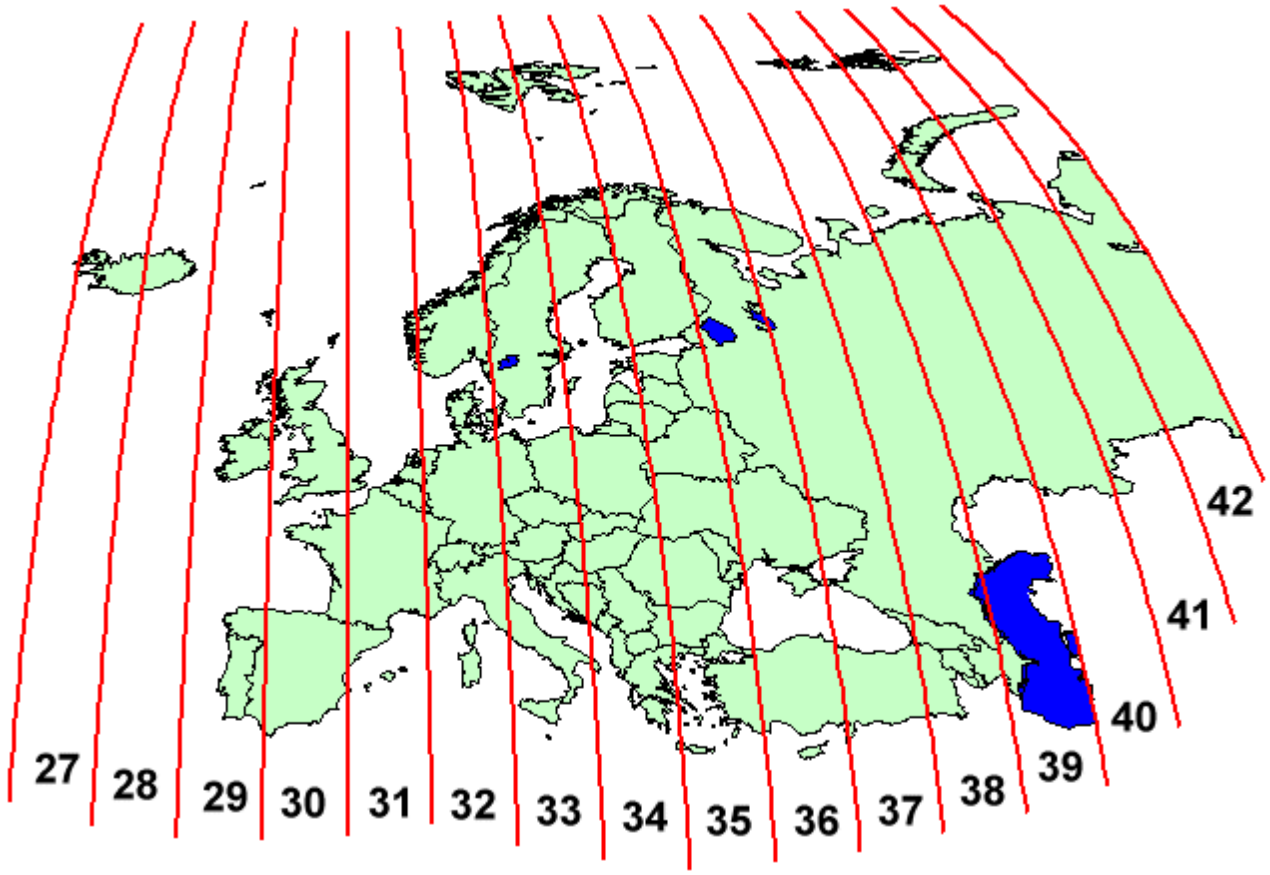
- ✓ The Earth is split into 60 vertical zones, numbered 1 to 60 from west to east.
- ✓ Each zone has a central meridian (0 distortion line) for accurate mapping.
- ✓ The system uses two hemispheres: North (N) and South (S).
- ✓ Coordinates are in meters (Easting & Northing), making calculations easier than degrees.



PROJECTED COORDINATE SYSTEM (PCS)

- **Example:**

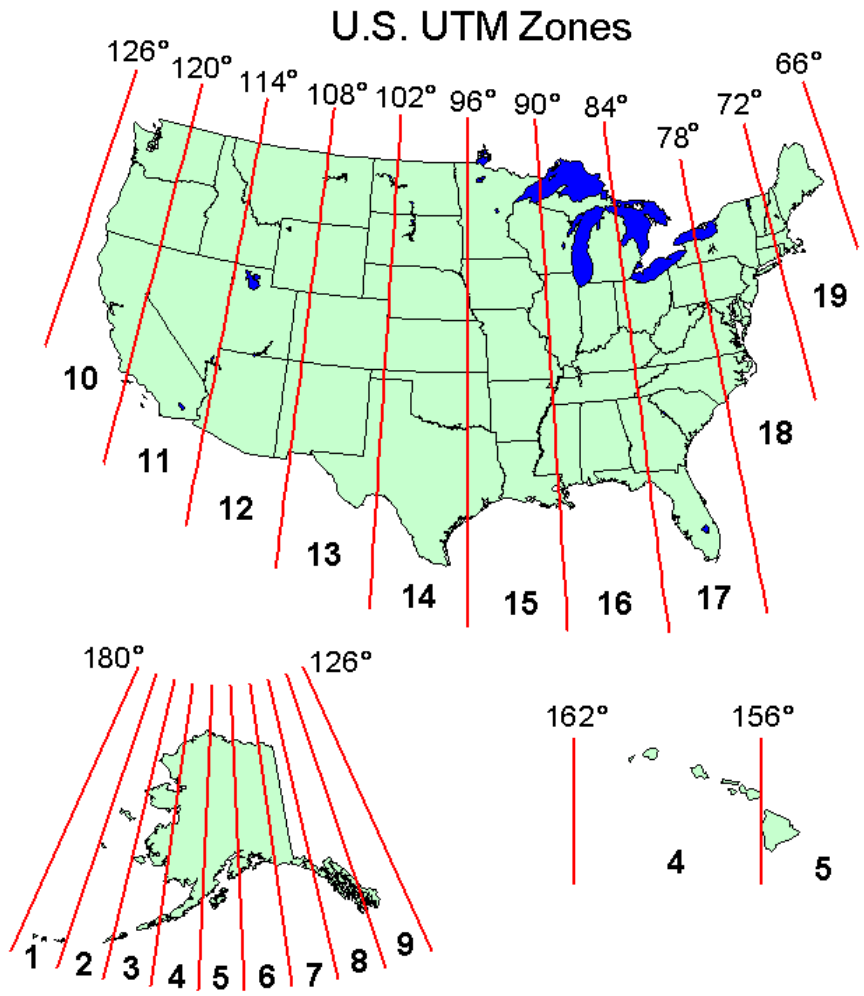
1. Paris, France
   - ✓ Latitude/Longitude: 48.8566° N, 2.3522° E
   - ✓ UTM Zone: 31N
   - ✓ UTM Coordinates: Easting: 448251 m, Northing: 5411932 m



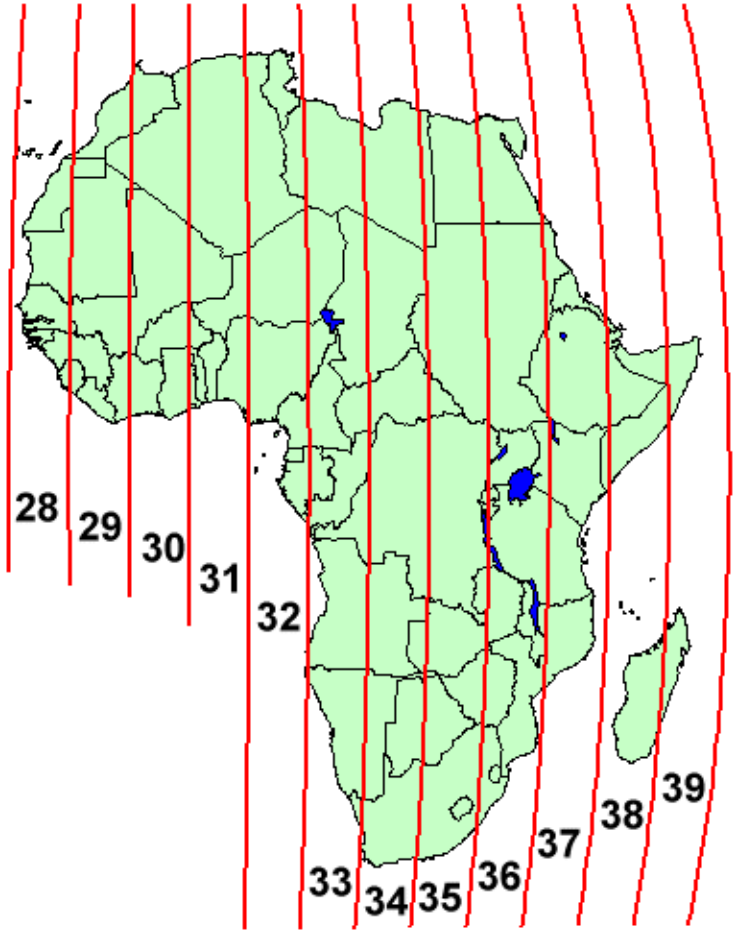
PROJECTED COORDINATE SYSTEM (PCS)

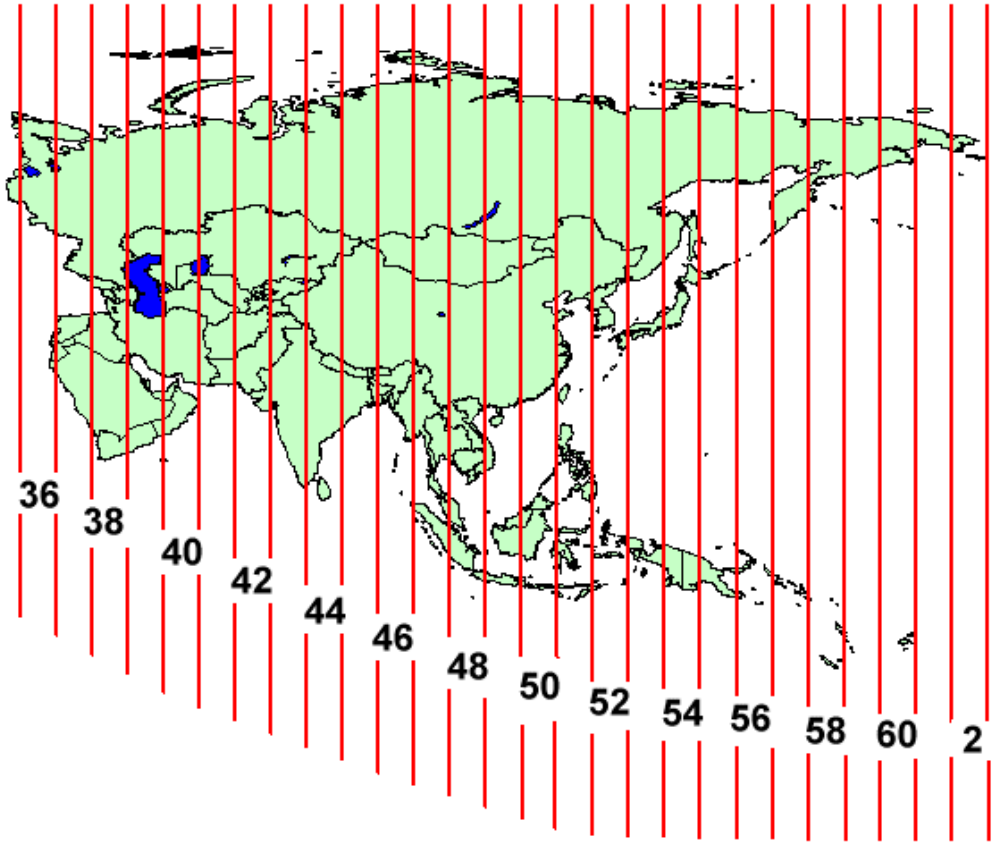
- **Example:**

1. New York City, USA
   - ✓ Latitude/Longitude: 40.7128° N, 74.0060° W
   - ✓ UTM Zone: 18N
   - ✓ UTM Coordinates: Easting: 585628 m, Northing: 4501045 m

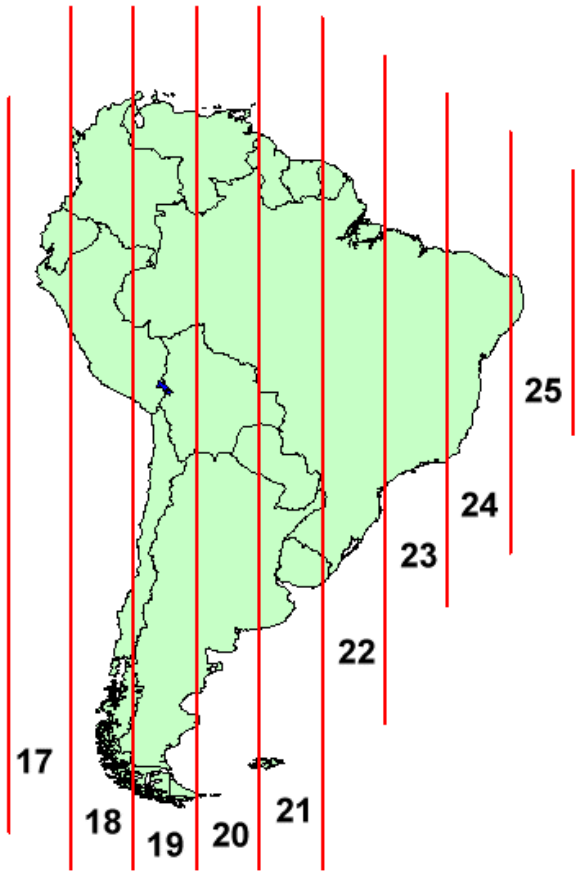


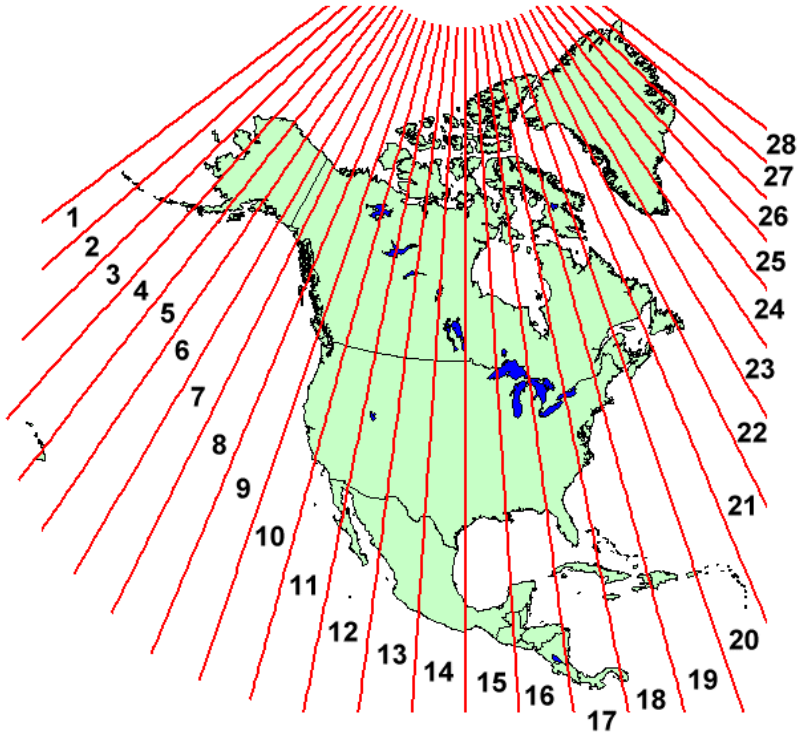
PROJECTED COORDINATE SYSTEM (PCS)



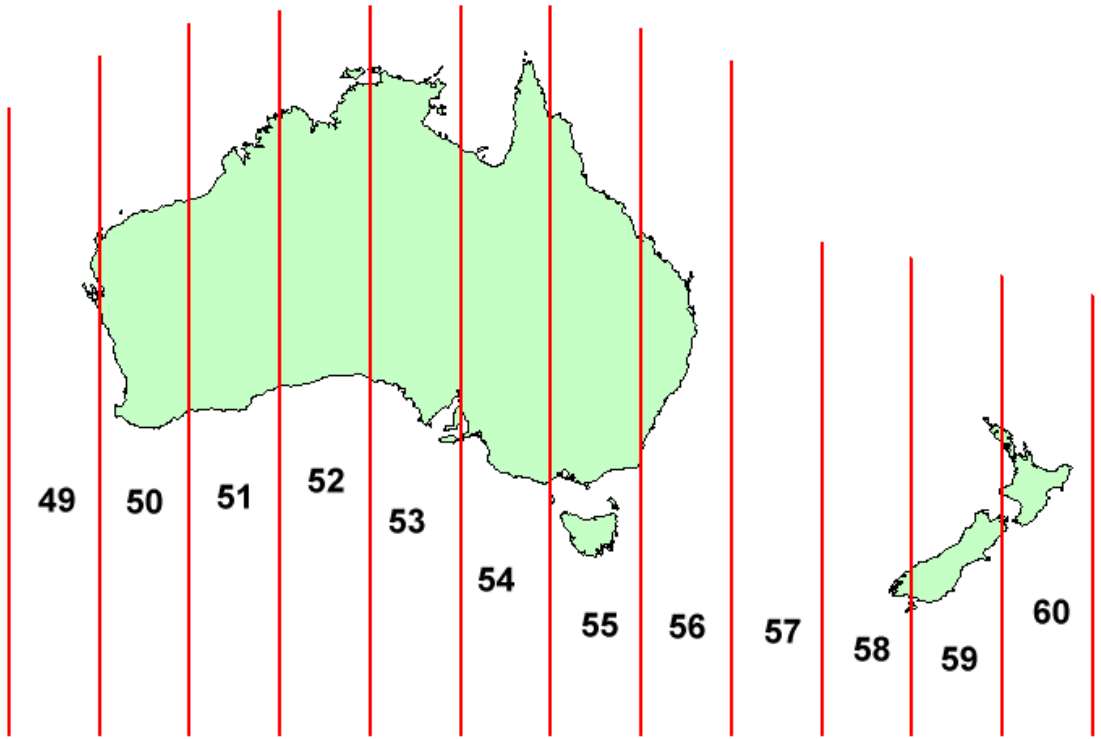


PROJECTED COORDINATE SYSTEM (PCS)





PROJECTED COORDINATE SYSTEM (PCS)



PROJECTED COORDINATE SYSTEM (PCS)

How to Find UTM Coordinates for Any Point

1- Use Online Converters like below websites:

- Convert Coordinates
- LatLong.net

2- Google Earth:

- Open Google Earth Pro
- Click on any location
- Look at the bottom of the screen for UTM coordinates

PROJECTED COORDINATE SYSTEM (PCS)

Why Use UTM Instead of Latitude/Longitude?

✅ Advantages of UTM:

✓ More accurate for local maps (less distortion than Mercator).

✓ Easier distance measurement (meters instead of degrees).

✓ Works well for GPS, GIS, and land surveys.

❌ Disadvantages of UTM:

✓ Zone boundaries can be tricky (two nearby points might be in different zones).

✓ Not ideal for global maps (only useful within one zone).

📌 Best for: Small to medium-scale maps (e.g., city planning, engineering, military mapping).

MAP SCALE AND MAP PROJECTIONS

Scale Distortions and Map Projections

- Why does scale change across maps?
  - ✓ The Earth is curved, but maps are flat.
  - ✓ Scale is accurate at the center but distorted at the edges.
  - ✓ Large-scale maps have less distortion than small-scale maps.
- Example:
  - ✓ Greenland looks huge on a world map due to distortion, but it's much smaller in reality.

MAP ABSTRACTION

- **How Geographic Information is Simplified for Mapping ?**
- Answer: Using map abstraction
- **What is Map Abstraction?**

✅ Map abstraction is the process of simplifying real-world geographic features to make them easier to represent on a map.

✅ Since the world is too complex to show everything, map abstraction removes unnecessary details while keeping important information.

✅ It involves selecting, simplifying, classifying, symbolizing, and representing data.

📌 Example: A world map doesn't show every house and street but keeps countries, cities, and major rivers.

MAP ABSTRACTION

Why is Map Abstraction Important?

📌 Maps must be simple and clear – Too much detail can make them hard to read.

📌 Different maps need different levels of detail – A city map needs streets, but a world map does not.

📌 Abstraction helps in GIS and cartography – It allows for accurate data visualization at different scales.

✅ Example: A road map shows only major highways at a small scale but includes all streets at a large scale.

MAP ABSTRACTION

The Five Stages of Map Abstraction

Map abstraction involves five key processes:

1. **Selection** – Choosing which real-world features to include.
2. **Simplification** – Removing unnecessary details from features.
3. **Classification** – Grouping similar features together.
4. **Symbolization** – Using symbols, colors, and lines to represent features.
5. **Representation** – Displaying the abstracted data in a meaningful way.

📌 Example: A river on a map is shown as a simple blue line, not as its real, complex shape.

MAP ABSTRACTION

Step 1: Selection – Choosing What to Show on a Map

What things affect our choice at this step?

MAP ABSTRACTION

Step 1: Selection – Choosing What to Show on a Map

✅ Not everything in the real world can fit on a map.

✅ Selection decides what features to keep, remove, or simplify based on the map's purpose.

✅ The map scale affects selection – Small-scale maps show fewer details, while large-scale maps show more.

📌 Example:

✓ A tourist map may show landmarks and roads but not small buildings.

✓ A weather map may show temperature zones but not city streets.

MAP ABSTRACTION





Here we have a tourist map (left) and a snow forecasting map (right) for New York.

You can see how different they are

MAP ABSTRACTION

Step 2: Simplification – Removing Unnecessary Details

✅ Why? Complex features are hard to read on a map.

✅ Simplification removes small details while keeping the main shape.

✅ Used for roads, rivers, coastlines, and city boundaries.

📌 Example:

✓ A winding river in reality is shown as a smooth curve on a map.

✓ A detailed mountain range is shown with basic contour lines.

MAP ABSTRACTION

Step 3: Classification – Grouping Similar Features

✅ Why ? Many different features can make a map confusing.

✅ Classification organizes data into categories to make it easier to understand.

📌 Example:

✓ Roads can be classified into highways, main roads, and small streets.

✓ Elevation levels can be grouped as low, medium, or high instead of showing every small change.

✅ Result: A map becomes simpler and clearer.

MAP ABSTRACTION

Step 4: Symbolization – Using Symbols to Represent Features

✅ Maps use symbols, colors, and lines instead of real-world images.

✅ Symbols help reduce clutter and improve readability.

✅ Different shapes, colors, and line styles represent different features.

📌 Examples of Symbolization:

- 🏙️ Cities → Dots or circles
- 🏔️ Mountains → Brown triangles or contour lines
- 🚗 Roads → Black or red lines
- 🌊 Water Bodies → Blue areas

✅ Symbols must be easy to recognize and consistent across maps.

MAP ABSTRACTION

Find some more symbols to showing other geographic features like green spaces, forest, residential areas, deserts and …

MAP ABSTRACTION

Step 5: Representation – Displaying the Data Effectively

✅ Representation is the final step, where all abstracted features are arranged in a logical, readable way.

✅ It ensures that the map serves its purpose, whether for navigation, analysis, or visualization.

📌 Example: A subway map represents stations and routes using simple lines and dots, even though subway routes in reality are complex.

✅ A good representation should be useful and easy-to-read map!

MAP ABSTRACTION

Draw a tourist map for your city's downtown

Don't need much detail

Do all the steps that mentioned before

Note: unlike mental maps this map should be understandable and should express everything by itself and don't need you to explain.

MAP ABSTRACTION

Example of Map Abstraction in Action

🔍 Imagine making a city map:

1. Selection: Keep main roads, rivers, and buildings; remove side streets and small objects.
2. Simplification: Show highways as straight lines, not with all their curves.
3. Classification: Group roads into highways, main roads, and side streets.
4. Symbolization: Use different colors and icons for roads, parks, and water bodies.
5. Representation: Arrange everything so the map is easy to read.

✅ Result: A clear, simple map that helps users find locations easily.

MAP ABSTRACTION

Real-World Applications of Map Abstraction

📌 Where is map abstraction used?

1. Navigation apps – Google Maps simplifies streets for easier viewing.
2. Weather maps – Group temperature zones instead of showing every small change.
3. Subway maps – Simplify train routes for quick understanding.
4. Military maps – Show only key strategic features, not all buildings.
5. Thematic maps – Population maps group data into color-coded regions.

✅ Every map you see uses abstraction!

MAP ABSTRACTION

Common Problems with Map Abstraction

❌ Too much simplification = Loss of important details.

❌ Poor classification = Confusing categories.

❌ Bad symbolization = Hard-to-read maps.

❌ Misrepresentation = Misleading or incorrect maps.

✅ A good map must **balance simplicity and accuracy!**

Chapter 3

Data, Information, and Where to Find Them

DATA AND INFORMATION

What is Data?

✅ Data refers to raw facts, numbers, or observations collected from the real world.

✅ Data by itself has no meaning until it is processed or organized.

✅ Types of GIS Data Sources:

1. Maps (paper or digital)
2. Satellite images
3. Sensor data (temperature, rainfall)
4. Survey measurements

📌 Example: A list of temperatures recorded in a city is data, but it becomes useful when analyzed to detect weather patterns.

DATA AND INFORMATION

What is Information?

✅ Information is processed, organized, or analyzed data that gives meaning and helps decision-making.

✅ Difference Between Data & Information:

Data	Information
Raw numbers, facts, symbols	Organized and meaningful
Example: 25°C, 30°C, 27°C	Example: "The average temperature this week is 27°C"
Needs processing	Helps in decision-making

📌 **Example:** GPS records a list of locations (data), but a navigation app analyzes and provides directions (information).

DATA AND INFORMATION

The Relationship Between Data and Information

📌 Data is the input, and information is the output.

📌 GIS converts raw geographic data into useful information for planning, navigation, and analysis.

📌 Steps:

1. Collect data (e.g., rainfall records).
2. Organize and process (calculate the average).
3. Turn it into useful information (predict drought conditions).

✅ Example: A GIS system analyzing traffic data to provide real-time congestion maps.

TYPES OF DATA IN GIS

Types of Data in GIS

GIS uses two main types of data:

1️⃣ Spatial Data (Geographic Data) – Information about where things are located.

✓ Example: City locations, rivers, roads.

✓ Used in maps and navigation systems.

2️⃣ Attribute Data (Descriptive Data) – Information about features in a location.

✓ Example: Population of a city, type of road (highway or street).

✓ Used for analysis and decision-making.

📌 Example: A GIS database storing earthquake locations (spatial) and their magnitude (attribute).

3️⃣ Meta data (data about data)

TYPES OF DATA IN GIS

Spatial Data – Location-Based Information

✅ Spatial data tells us where something is on Earth.

✅ Represented using coordinates, maps, and GIS layers.

📌 Types of Spatial Data:

- ✓ Point Data: Single locations (e.g., schools, hospitals).
- ✓ Line Data: Roads, rivers, pipelines.
- ✓ Polygon Data: Areas (e.g., lakes, city boundaries, forests).

✅ Example: Google Maps uses spatial data to show locations of restaurants.

TYPES OF DATA IN GIS

Attribute Data – Descriptive Information

✅ Attribute data describes characteristics of spatial data.

✅ Stored in tables linked to maps.

📌 Examples of Attribute Data:

✓ A city has: Name, population, area.

✓ A road has: Type (highway or street), width, speed limit.

✓ A river has: Name, length, water quality.

✅ Example: GIS stores earthquake locations (spatial data) and their magnitude, depth, and date (attribute data).

PRACTICE

draw an earthquake map for your country which shows at least 10 earthquakes

it's going to be an imaginary map which shows spatial data, attribute data and metadata

TYPES OF DATA IN GIS

Difference Between Spatial and Attribute Data

Spatial Data	Attribute Data
Answers **"Where?"**	Answers **"What?"**
Example: Location of a school	Example: Number of students in the school
Shown on a map	Stored in a table

📌 Example: A GIS system for disease tracking:

✓ Spatial data: Shows where cases are reported.

✓ Attribute data: Includes patient age, symptoms, recovery rate.

TYPES OF DATA IN GIS

What is Metadata?

✅ Metadata = "Data about Data“

✅ It describes:

- ✓ What the data represents
- ✓ Where the data comes from
- ✓ How the data was collected
- ✓ When the data was created
- • Who created the data

📌 Example:

A satellite image file might have metadata like:

✓ Date captured: March 2024
✓ Resolution: 10 meters per pixel
✓ Source: Sentinel-2 satellite
✓ Projection system: WGS 84

TYPES OF DATA IN GIS

Why is Metadata Important in GIS?

📌 Metadata helps with:

✓ Understanding the data – Avoids misinterpretation.

✓ Finding the right data – Easier search and retrieval.

✓ Ensuring accuracy – Checks for errors.

✓ Data sharing – Helps others use the data correctly.

✅ Example: If two maps show different road networks, metadata tells us which is the most recent and accurate one.

TYPES OF DATA IN GIS

Key Elements of GIS Metadata

📌 Metadata typically includes:

✓ Title: What the dataset is called.

✓ Description: Brief summary of the data.

✓ Creator: Who made the data.

✓ Date: When the data was collected or updated.

✓ Source: Where the data comes from.

✓ Format: File type (Shapefile, GeoTIFF, CSV).

✓ Projection & Coordinate System: Geographic reference.

✓ Data Quality: Accuracy and limitations.

✅ Example: A Landsat satellite image will have metadata specifying when and where the image was captured and its spatial resolution.

TYPES OF DATA IN GIS

Example Metadata for a Digital Elevation Model (DEM)

Metadata Field	Example Value
Title	DEM of California (30m resolution)
Creator	USGS
Date Collected	February 2023
Data Source	LIDAR & Satellite Data
Format	GeoTIFF
Spatial Resolution	30 meters
Coordinate System	UTM Zone 10N, WGS 84

SOURCES OF GIS DATA

Sources of GIS Data

GIS collects data from many sources, including:

✅ Satellite Images – Remote sensing data (e.g., land use, weather).

✅ Field Surveys – Data collected manually using GPS.

✅ Government Databases – Census, climate records, infrastructure maps.

✅ Drones & Aerial Photography – High-resolution mapping.

📌 Example: GIS combines satellite images and population data to track urban growth.

CONVERTING GIS DATA

How GIS Converts Data into Information

1️⃣ Collect raw data – Sensor readings, GPS locations, surveys.

2️⃣ Process & organize – Convert into tables, graphs, and GIS layers.

3️⃣ Analyze patterns – Identify trends in climate, transportation, health, etc.

4️⃣ Create maps & reports – Convert into readable information.

✅ Example: GIS uses weather station data to generate storm prediction maps.

DATA QUALITY

Why Data Quality is Important in GIS

📌 Good data = Accurate information

✓ Errors in data lead to wrong conclusions and poor decisions.

✓ GIS must ensure data accuracy, consistency, and completeness.

❌ Problems with Poor Data:

✓ Incorrect GPS locations can mislead navigation apps.

✓ Outdated maps can cause wrong urban planning decisions.

✓ Low-quality satellite images make land use analysis difficult.

✅ Solution: Regularly update and validate GIS databases.

GIS DATA

Real-World Applications of GIS Data

📌 How GIS data is used in different fields:

✅ Urban Planning: Mapping city growth, zoning laws.

✅ Environmental Science: Monitoring deforestation, pollution.

✅ Disaster Management: Tracking earthquakes, floods, hurricanes.

✅ Agriculture: Precision farming, crop health analysis.

✅ Business & Marketing: Finding the best locations for new stores.

📌 Example: Google Maps analyzes traffic data to recommend the fastest route.

GIS DATA

Challenges in Collecting and Using GIS Data

❌ **Data collection is expensive** – Satellites, surveys, and sensors require funding.

❌ **Data inconsistency** – Different sources may provide conflicting information.

❌ **Privacy concerns** – GPS and personal location data can be misused.

❌ **Data overload** – Large GIS databases require powerful computers.

✅ **Solution:** Improve **data integration, validation, and security**.

GIS DATA

Class Activity – Exploring GIS Data

💡 **Activity:**

- Use **Google Earth or an online GIS tool**.
- Find a **famous landmark** (e.g., Eiffel Tower, Grand Canyon).
- Identify its **spatial data** and **attribute data.**
- Discuss: **How would you use this data for a project?**

FINDING GIS DATA

Key Questions Before Finding GIS Data

📌 Before searching for GIS data, ask:

1. What kind of data do I need? – Roads, elevation, population?
2. What format should it be in? – Shapefile, GeoTIFF, CSV?
3. What is the required spatial resolution? – High-resolution (detailed) or low-resolution?
4. Is the data up-to-date? – Old data may be unreliable.
5. Is the data free or requires payment?

✅ Example: A weather researcher needs recent rainfall data in raster format for analysis.

FINDING GIS DATA

Types of GIS Data Sources

📌 GIS data can be found in four main sources:

1️⃣ Government Agencies
✓ Example: USGS, NASA, European Space Agency (ESA).

2️⃣ Research Institutions & Universities
✓ Example: Harvard Geospatial Library, Earth Explorer.

3️⃣ Private Companies
✓ Example: Google Earth Engine, ESRI, Maxar Technologies.

4️⃣ Crowdsourced & Open Data Platforms
✓ Example: OpenStreetMap (OSM), Global Forest Watch.

FINDING GIS DATA

Government & Official Data Sources

📌 Many governments provide free GIS data for public use:

✓ United States Geological Survey (USGS) – Elevation, topography, and satellite imagery.

✓ NASA Earthdata – Satellite images, climate data.

✓ European Space Agency (ESA) – Sentinel satellite data.

✓ Natural Resources Canada (NRCan) – Topographic and environmental data.

✓ UK Ordnance Survey – Detailed mapping data of the UK.

✅ Example: If you need land cover maps, NASA's MODIS Land Cover dataset is a good source.

FINDING GIS DATA

Research & University GIS Data Sources

📌 Universities and research centers provide scientific GIS datasets:

✓ Harvard Geospatial Library – Historical maps and spatial datasets.

✓ NOAA (National Oceanic and Atmospheric Administration) – Climate and ocean data.

✓ FAO GeoNetwork – Agricultural and land cover data.

✓ Earth Explorer (by USGS) – Satellite imagery and remote sensing data.

✓ World Bank Open Data – Economic and demographic datasets.

✅ Example: A researcher studying deforestation can use FAO GeoNetwork for forest cover data.

FINDING GIS DATA

Private & Commercial GIS Data Providers

📌 Some companies offer high-resolution GIS data (often for a fee):

✓ ESRI ArcGIS Hub – GIS datasets from various sources.

✓ Google Earth Engine – Satellite imagery & cloud-based analysis.

✓ Maxar Technologies – Very high-resolution satellite images.

✓ Airbus OneAtlas – Commercial satellite and aerial imagery.

✓ HERE Technologies – Transportation and road network data.

✅ Example: If a company needs high-accuracy building height data, they might purchase Maxar satellite images.

FINDING GIS DATA

Free & Crowdsourced GIS Data Sources

📌 Open-source GIS data is created by communities and volunteers:

✓ OpenStreetMap (OSM) – Free global map data (roads, buildings, etc.).

✓ Global Forest Watch – Deforestation and forest monitoring data.

✓ Humanitarian Data Exchange (HDX) – GIS data for disaster response.

✓ Natural Earth – Free vector and raster data for cartography.

✓ Google Open Data Portal – Public datasets from various sources.

✅ Example: A student mapping urban areas can use OSM building footprints for free.

FINDING GIS DATA

Finding GIS Data Using Online Portals

📌 Where to search for GIS data?

✓ Google Dataset Search – Helps find GIS datasets across the web.

✓ ArcGIS Living Atlas – A vast collection of GIS-ready datasets.

✓ Copernicus Open Hub – Free Sentinel satellite data.

✓ USGS Earth Explorer – A search tool for satellite images.

✓ Data.gov – Open government datasets (US).

✅ Example: A climate scientist looking for temperature trends can search Copernicus Open Hub for free climate data.

FINDING GIS DATA

How to Choose the Right GIS Dataset?

📌 When selecting a dataset, check:

✓ Accuracy & Reliability – Who created it? Is it trusted?

✓ Date of Collection – Is it recent?

✓ Spatial Resolution – Is it detailed enough?

✓ Projection & Coordinate System – Does it match your project?

✓ License & Restrictions – Can you use it for commercial or academic purposes?

✅ Example: A disaster management team chooses the latest satellite images to assess flood damage.

FINDING GIS DATA

How to Download & Use GIS Data?

📌 Steps to obtain and use GIS data:

1️⃣ Search for data – Use portals like USGS, NASA, OpenStreetMap.

2️⃣ Check metadata – Ensure it matches project needs.

3️⃣ Download in the correct format – Shapefile, GeoTIFF, CSV, etc.

4️⃣ Load into GIS software – ArcGIS, QGIS, Google Earth.

5️⃣ Perform analysis – Use GIS tools for visualization and study.

✅ Example: A GIS analyst downloads Landsat images from Earth Explorer, then analyzes vegetation changes in QGIS.

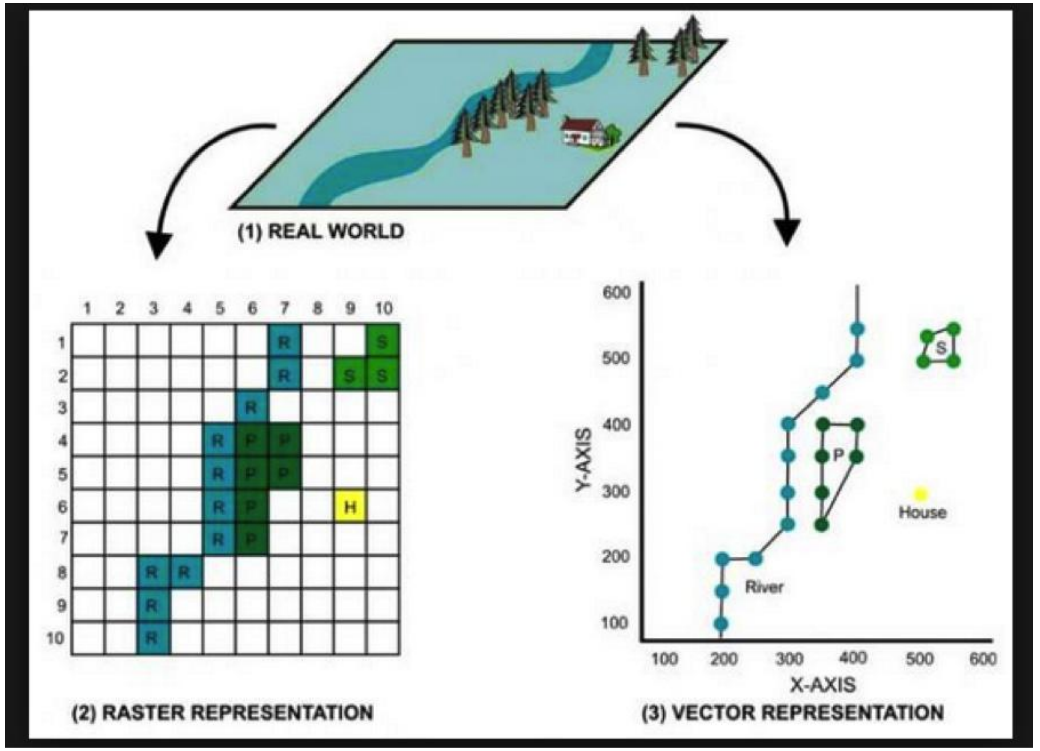
Chapter 4

Data models in GIS

DATA MODELS IN GIS

Raster Data vs. Vector Data in GIS

- In Geographic Information Systems (GIS), Raster Data and Vector Data are the two main types of spatial data models used to represent geographic features.
- Both have unique characteristics, advantages, and applications.



DATA MODELS IN GIS

Raster Data

Raster data represents the world as a grid of cells (pixels) where each cell has a value representing an attribute such as elevation, temperature, or land cover.

📌 Key Characteristics:

✓ Made up of rows and columns (grid structure).

✓ Each cell (pixel) has a value that represents a feature.

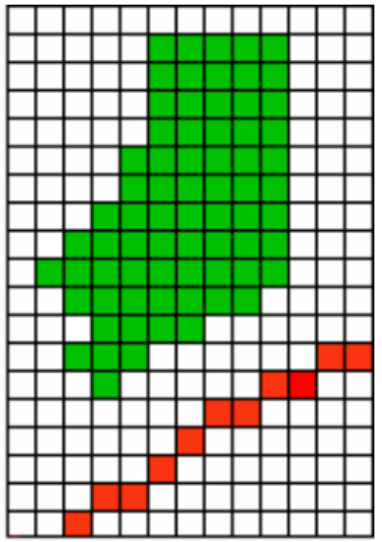
✓ Can store continuous data (e.g., elevation, temperature).

✓ Resolution depends on pixel size (smaller pixels = higher resolution).

✅ Example: A satellite image where each pixel stores a color or spectral value.



Real World



Raster

DATA MODELS IN GIS

Types of Raster Data:

1. Satellite Images & Aerial Photos

📌 Example: Landsat, Sentinel, and Google Earth images.
✓ Used for land cover classification, vegetation analysis, and environmental monitoring.

2. Digital Elevation Models (DEM)

📌 Example: SRTM, ASTER DEM.
✓ Represents terrain elevation in a grid format.
✓ Used for topographic mapping, flood modeling, and slope analysis.

3. Thematic Raster Data

📌 Example: Land use maps, soil type maps.
✓ Each pixel has a value representing a category (e.g., forest, water, urban).
✓ Used in land cover analysis and urban planning.

4. Continuous Raster Data

📌 Example: Temperature, precipitation, ocean salinity.
✓ Values change continuously over space.
✓ Used in climate modeling and hydrology studies.

DATA MODELS IN GIS

Raster Data Formats:

✓ TIFF (.tif) – Common high-quality format.

✓ JPEG (.jpg), PNG (.png) – Compressed images.

✓ GRID – ESRI-specific raster format.

✓ NetCDF – Scientific raster data format.

DATA MODELS IN GIS

Advantages of Raster Data:

- ✅ Ideal for continuous data (e.g., elevation, temperature).
- ✅ Easy to process for remote sensing and satellite imagery.
- ✅ Works well with raster-based spatial analysis (e.g., slope, aspect, hydrology).
- ✅ Compatible with machine learning and AI models.

DATA MODELS IN GIS

Disadvantages of Raster Data:

- ❌ Large file sizes for high-resolution images.
- ❌ Limited attribute storage (only one value per cell).
- ❌ Loses detail when zoomed in (pixelation).
- ❌ Complex spatial analysis requires heavy computing power.

DATA MODELS IN GIS

Vector Data

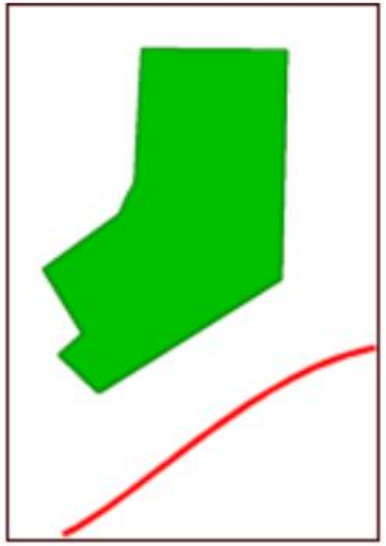
Vector data represents geographic features using points, lines, and polygons. It is best for discrete objects such as roads, buildings, and political boundaries.

📌 Key Characteristics:

✓ Uses mathematical coordinates (X, Y) instead of pixels.
✓ Stores discrete features like cities, roads, and land parcels.
✓ More precise than raster data.
✓ Supports multiple attributes per feature.



Real World



Vector

✅ Example: A road network represented by lines with attributes like name, speed limit, and road type.

DATA MODELS IN GIS

Types of Vector Data:

1. Point Data (0D)

📌 Represents single locations using X, Y coordinates.
✓ Used for cities, trees, wells, weather stations.
✓ Example: A dataset of all hospitals in a city.

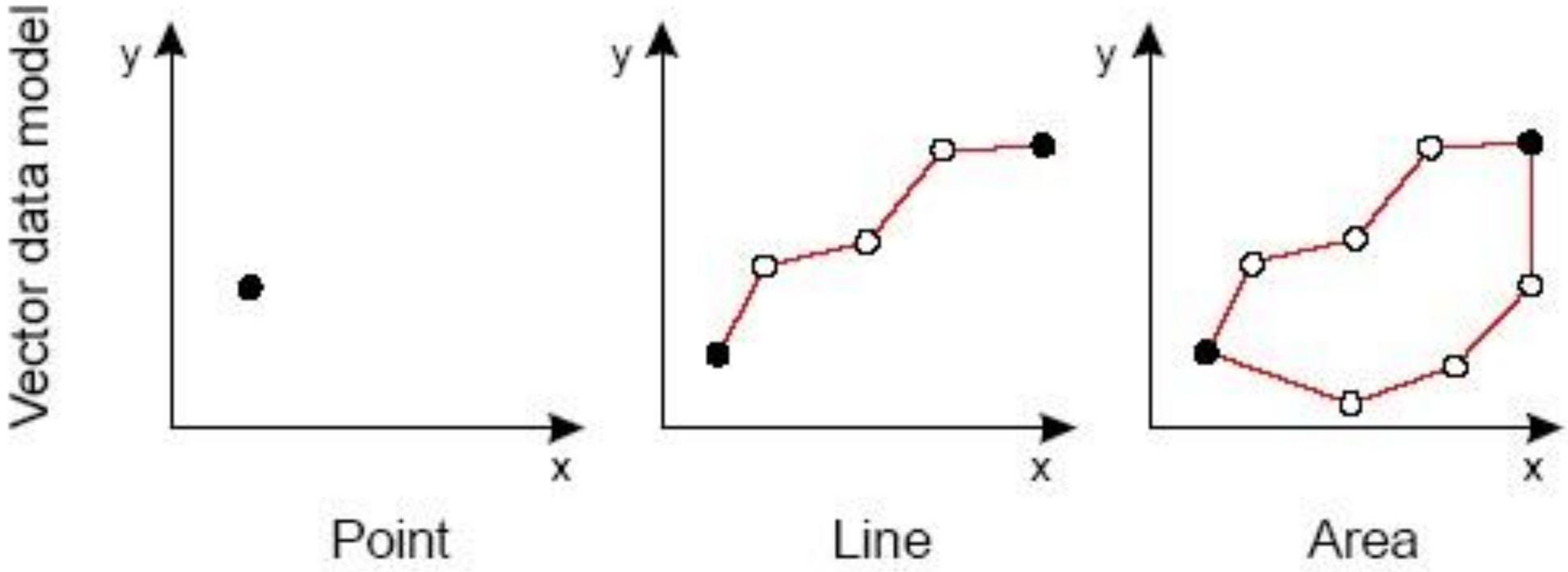
2. Line Data (1D)

📌 Represents linear features connecting points.
✓ Used for roads, rivers, pipelines, power lines.
✓ Example: A transportation network of highways.

3. Polygon Data (2D)

- 📌 Represents areas using connected lines forming a closed shape.
  ✓ Used for countries, lakes, land use zones.
  ✓ Example: A shapefile of administrative boundaries.

DATA MODELS IN GIS



DATA MODELS IN GIS

Vector Data Formats:

✓ Shapefile (.shp) – Common GIS format for vector data.

✓ GeoJSON (.geojson) – Web-friendly GIS format.

✓ KML (.kml) – Used for Google Earth.

✓ GDB (Geodatabase) – Advanced ESRI database format.

DATA MODELS IN GIS

Advantages of Vector Data:

- ✅ More precise than raster data.
- ✅ Supports multiple attributes per feature.
- ✅ File sizes are smaller compared to high-resolution raster data.
- ✅ Best for map-making and cartography.
- ✅ Allows topological relationships (e.g., which roads intersect).

DATA MODELS IN GIS

Disadvantages of Vector Data:

- ❌ Not ideal for continuous data like elevation or temperature.
- ❌ Complex geometries may slow down analysis.
- ❌ Converting from raster to vector is time-consuming and may lose detail.

DATA MODELS IN GIS

Raster vs. Vector – Key Differences

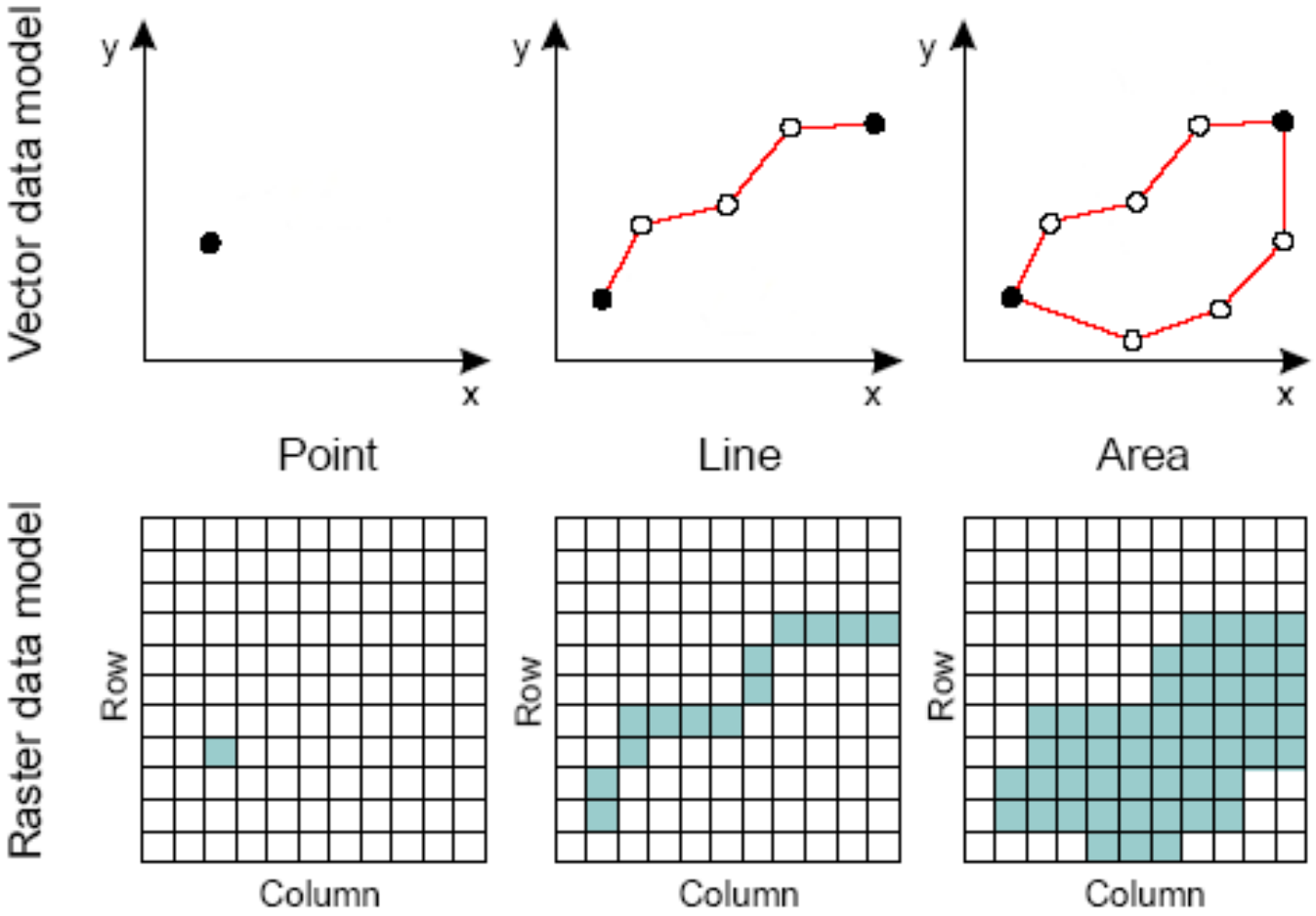
Feature	Raster Data	Vector Data
Structure	Grid (cells/pixels)	Points, lines, polygons
Best for	Continuous data (elevation, temperature)	Discrete objects (roads, boundaries)
Storage	One value per cell	Multiple attributes per feature
File Size	Large for high resolution	Smaller than raster
Scalability	Loses detail when zoomed	Maintains detail
Computing Power	Requires more processing	Easier to process

✅ **Example:**

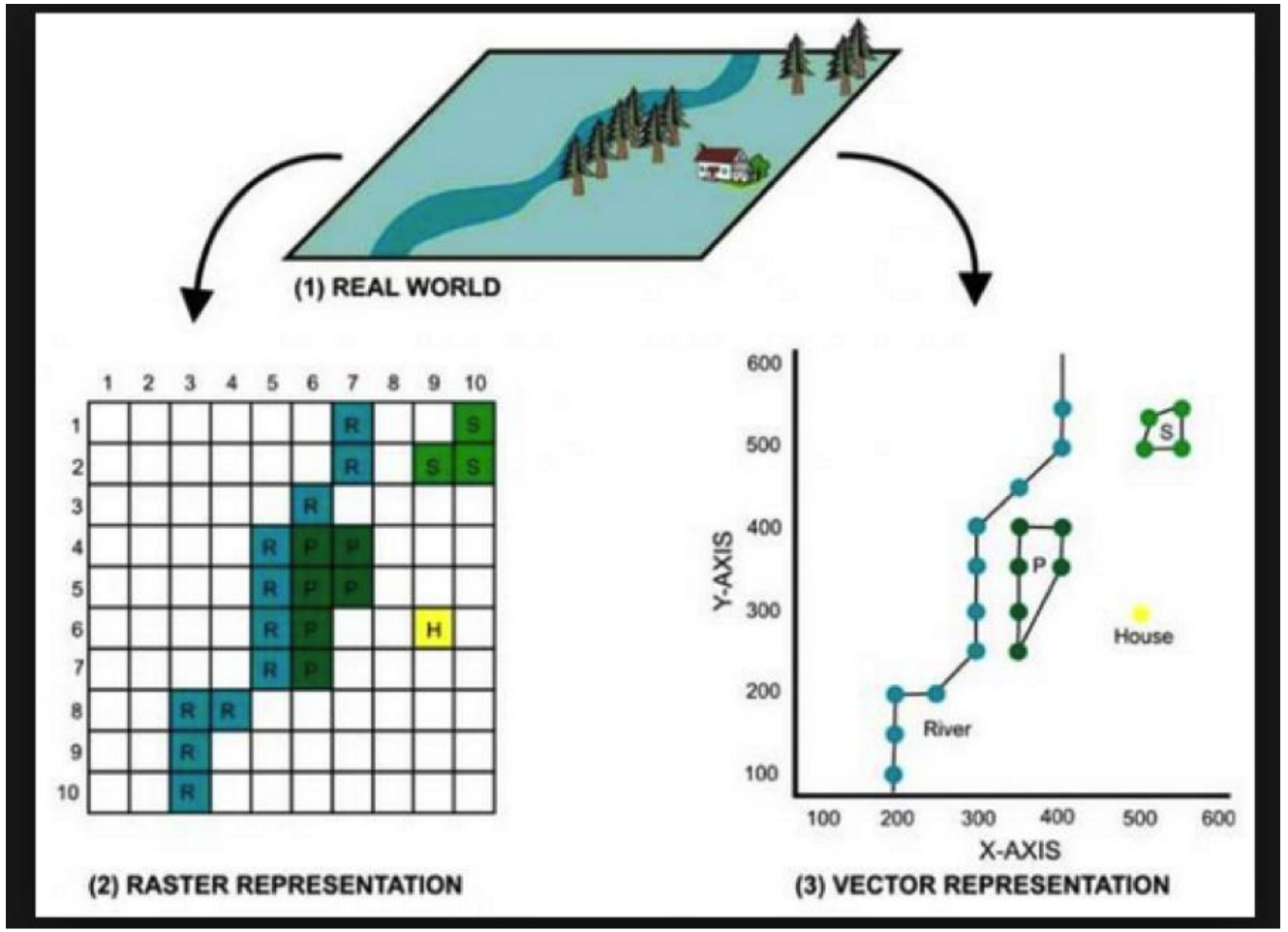
✔ **Use Raster** for **satellite imagery** or **climate data**.

✔ **Use Vector** for **road networks, administrative boundaries**.

DATA MODELS IN GIS



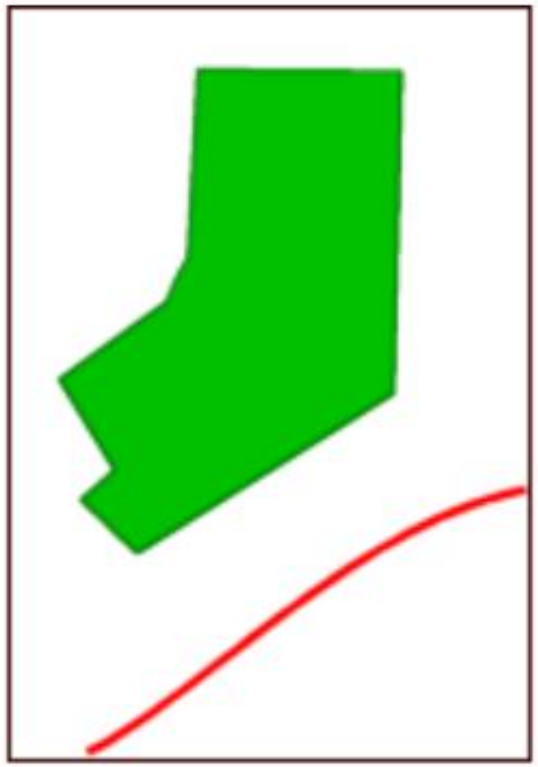
DATA MODELS IN GIS



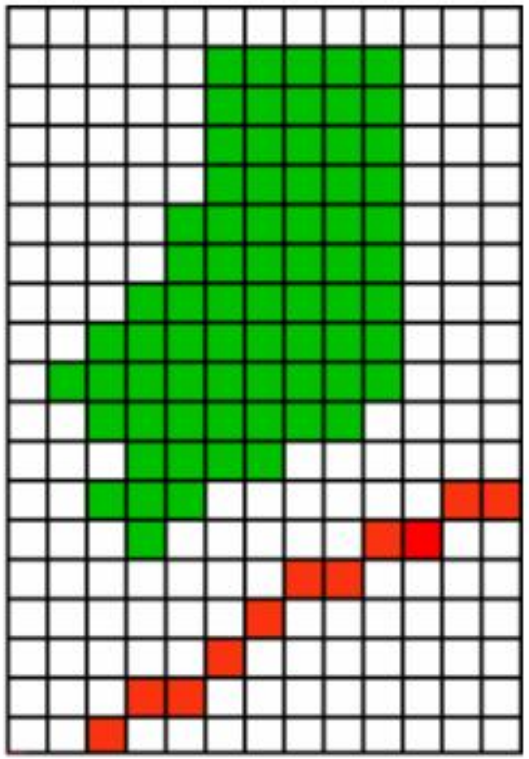
DATA MODELS IN GIS



Real World

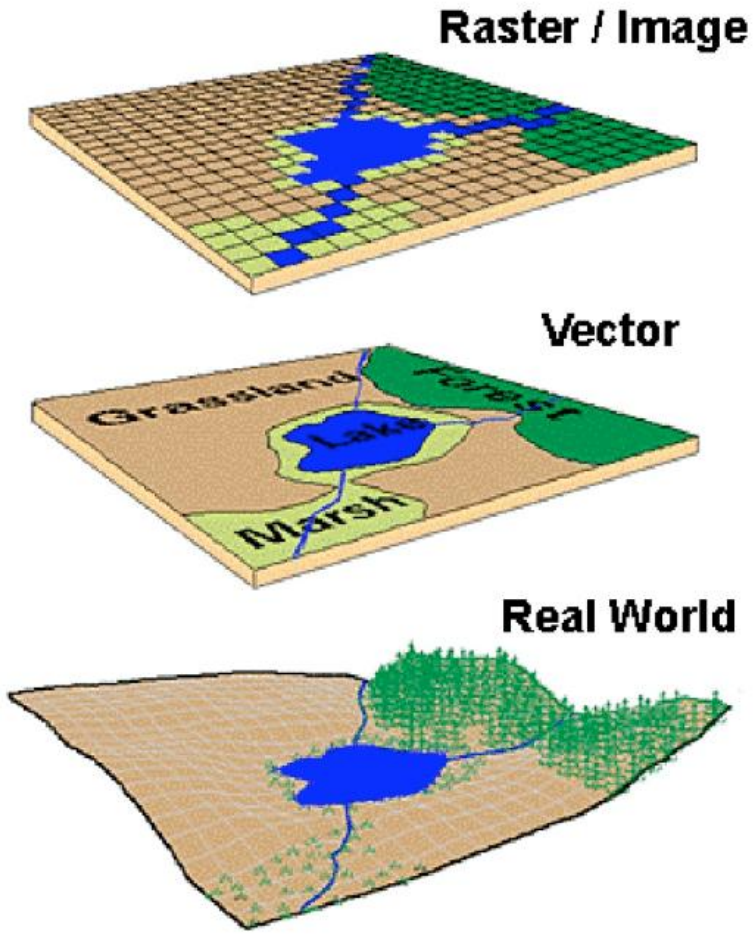


Vector



Raster

DATA MODELS IN GIS



DATA MODELS IN GIS

Converting Between Raster and Vector

Sometimes, we need to convert data between raster and vector:

- Raster to Vector (Vectorization) – Converts raster cells into polygons or points.
- Used in land cover classification to create vector maps.

- Vector to Raster (Rasterization) – Converts vector data into a grid of pixels.
- Used in GIS modeling where raster tools are needed.

Example: A forest cover raster is converted into vector polygons to measure the area of deforestation.

DATA MODELS IN GIS

Applications of Raster and Vector Data

- Raster Data Uses
  - Remote Sensing: Satellite images, land cover classification.
  - Climate Studies: Rainfall, temperature mapping.
  - Elevation Models: Digital Elevation Models (DEM).
  - Environmental Monitoring: Deforestation tracking, fire mapping.
- Vector Data Uses
  - Navigation & Transportation: Roads, railways, flight paths.
  - Urban Planning: Building footprints, zoning maps.
  - Cadastral Mapping: Land ownership boundaries.
  - Hydrology Studies: Rivers, watershed boundaries.

DATA MODELS IN GIS

Which One to Use?

📌 Use Raster Data if:

- ✓ You need satellite imagery or elevation models.
- ✓ Your data is continuous (e.g., temperature, pollution).
- ✓ You need to perform raster-based analysis (slope, aspect).

📌 Use Vector Data if:

- ✓ You need precise location data (cities, roads).
- ✓ Your data is categorical (e.g., land use types).
- ✓ You need small file sizes with rich attribute information.